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MCY 611
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Cloud Computing Project
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MCY 611 Cloud Computing Project

In this project, you will measure and compare the performance of a static web page hosted in S3 with a web page hosted on an EC2 instance. In Lab 1, you have learned AWS CLI and EC2 instances. Also, you have learned how to host a website on an EC2 instance. In Lab 2, you have learned S3 and how to host a web page (a static website) in S3. In Lab 4, you have learned an Apache HTTP server benchmarking tool, ab. You will use all you have learned to complete this project.

Requirements

1. The static web page you need for this project is as follows:

```
<html><body><h1>It works!</h1>
<p>This website is hosted by Amazon AWS</p>
</body></html>
```

You can name it index.html.

2. You need to host this web page on an EC2 instance and in S3.

3. You need to use the ab command to conduct performance tests. For the EC2 hosting, you need to conduct five ab tests on the web page with 50, 100, 150, 200, and 250 simultaneous requests. The number of requests to perform for each test is 5000. The five ab test commands are as follows:

```
ab -c 50 -n 5000 http://18.215.239.60/
ab -c 100 -n 5000 http://18.215.239.60/
ab -c 150 -n 5000 http://18.215.239.60/
ab -c 200 -n 5000 http://18.215.239.60/
ab -c 250 -n 5000 http://18.215.239.60/
```

Please look at <https://httpd.apache.org/docs/2.4/programs/ab.html> for the description of the ab command output. The “Time per request” value is the average time spent per request. There are two “Time per request” values in the output. You need to record the first value for each test.

For the S3 hosting, you also need to conduct five ab tests on the web page with 50, 100, 150, 200, and 250 simultaneous requests. The number of requests to perform for each test is 5000.

The five ab test commands are as follows:

```
ab -c 50 -n 5000 https://carrd12.s3.amazonaws.com/index.html/
ab -c 100 -n 5000 https://carrd12.s3.amazonaws.com/index.html/
ab -c 150 -n 5000 https://carrd12.s3.amazonaws.com/index.html/
ab -c 200 -n 5000 https://carrd12.s3.amazonaws.com/index.html/
ab -c 250 -n 5000 https://carrd12.s3.amazonaws.com/index.html/
```

You need to record the “Time per request” value (the first value) for each test.

You need to store data obtained from your tests in a table. Based on the data, you need to plot a graph to compare the performance of the index.html page hosted in S3 with the index.html page hosted on an EC2 instance at different numbers of simultaneous requests. Also, you need to analyze the graph.

4. Using the AWS Pricing Calculator at <https://calculator.aws/> to estimate the cost of hosting a static website in S3 and EC2 for a period of one month. You can make an assumption on the monthly usage of your static website so that you can estimate the cost.

Report

For this project, you need to turn in a report in a word document to the Canvas. The report must include all the sections listed below. Missing a section title, except for the title page, will be treated as if the section is missing.

1. **Title Page (5 points)** containing the title of the project, your name, the name of the class, your instructor's name, and the date the project was submitted.
2. **Introduction (5 points)** A single paragraph that describes the objectives or purpose of the project.
3. **Materials (5 points)** List all materials used in the project, including AWS CLI version, EC2 image ID, VPC ID, Subnet ID, Security credentials (your AWS Access Key ID and your AWS Secret Access Key), and S3 buckets. (Note: you only need to show the last five characters of your AWS Access Key ID and the last five characters of your AWS Secret Access Key. For example, the AWS Access Key ID is *****AMPLE.).
4. **Methods (45 points)** Describe the steps completed during your investigation. Be sufficiently detailed so that anyone could read this section and duplicate your efforts. You must use AWS CLI commands to complete this section. Using any non-CLI commands, such as the Management Console, will result in a score of 0 for the project. You must provide all your complete CLI commands along with the complete outputs. Any incomplete command or incomplete output will be treated as a wrong step. You cannot reuse any AWS resources (such as a security group, a key pair, etc.) from the labs, except for the access key and secret access key created in Lab 1, as well as your VPC ID and Subnet ID. You must include the AWS CLI commands you used to find your VPC ID and Subnet ID. Your security group, key pair, and S3 bucket names must start with your NKU username. Failure to follow this naming convention will result in a score of 0 for the section.
5. **Data (10 points)** Numerical data obtained from your procedure presented as a table. Data encompasses everything recorded when you conducted the experiment, without any interpretation of what they mean.
6. **Performance Analysis (10 points)** Describe in words and a graph what the data means. This is the section where you interpret the data. Label your graph with a descriptive title and label the axes, including units of measurement.
7. **Economic Analysis (10 points)** Describe the cost differences between the two solutions using both a written explanation and a graph. Ensure your graph has a descriptive title and clearly labels the two solutions along with their respective costs. Your analysis should include answers to the questions: Which solution for static website hosting is cheaper? Is the cheaper alternative faster or slower than the more expensive solution?
8. **Conclusion (5 points)** A short summary of the project, including its purpose and the meaning of your results. Advise the reader on which web hosting solution to use.
9. **References (5 points)** List any references that you used during the course of your project.

Introduction

This project's goal is to evaluate and contrast the performance of a static webpage hosted on an EC2 instance to a webpage maintained on Amazon S3. I obtained practical experience using a variety of AWS services and tools during the assignment. Lab 1 taught me how to host a website on an EC2 instance, set up EC2 instances, and utilize the AWS CLI. I successfully hosted a static website in Lab 2 after investigating Amazon S3. In Lab 4, I discovered how to evaluate web performance using the Apache HTTP server benchmarking tool `ab -V`. I was able to evaluate the performance variations between the two hosting solutions by putting the knowledge I had learned in these labs to use.

Tools and Materials

1.AWS Cloudshell terminal

2.AWS CLI

3.AWS Version

```
aws-cli/2.23.13 Python/3.12.6 Linux/6.1.127-135.201.amzn2023.x86_64 exec-env/CloudShell exe/x86_64.amzn.2023
```

4.AWS Configure

AWS Access Key ID [*****34RC]

AWS Secret Access Key [*****aLPU]

Default region name [us-east-1]

Default output format [json]

5.VPC ID

vpcId: vpc-0e2544332a068204f

6.Subnet ID

subnet-0aa6a451bcda4099f

7.Ubuntu Server 20.04 LTS AMI: ami-09e67e426f25ce0d7

8.Key pair File

9.ab -V tool

Methods

EC2

Verify if the CLI is installed properly via the following command:

```
~ $ aws --version
aws-cli/2.23.13 Python/3.12.6 Linux/6.1.127-135.201.amzn2023.x86_64 exec-env/CloudShell exe/x86_64.amzn.2023
```

Configure the AWS CLI with the `aws_cli_user` account (aws configure):

```
~ $ aws configure
AWS Access Key ID [None]: AKIA42PHHZCTYBP434RC
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
```

Get VpcId:

```

~ $ aws ec2 describe-vpcs --filters Name=is-default,Values=true
{
  "Vpcs": [
    {
      "OwnerId": "881490118823",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0688d6164b79f37c6",
          "CidrBlock": "172.31.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": true,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-0e2544332a068204f",
      "State": "available",
      "CidrBlock": "172.31.0.0/16",
      "DhcpOptionsId": "dopt-08b45b45933d7afb0"
    }
  ]
}

```

Get SubnetId:

```

~ $ aws ec2 describe-subnets --filters Name=vpc-id,Values=vpc-0e2544332a068204f Name=availability-zone,Values=us-east-1a Name=default-for-az,Values=true
{
  "Subnets": [
    {
      "AvailabilityZoneId": "us-east-1a",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "881490118823",
      "AssignIpv6AddressesOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:us-east-1:881490118823:subnet/subnet-0aa6a451bcd4099f",
      "EnableDns64": false,
      "IpV6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-0aa6a451bcd4099f",
      "State": "available",
      "VpcId": "vpc-0e2544332a068204f",
      "CidrBlock": "172.31.32.0/20",
      "AvailableIpAddressCount": 4091,
      "AvailabilityZone": "us-east-1a",
      "DefaultForAz": true,
      "MapPublicIpOnLaunch": true
    }
  ]
}

```

Create security group:

```

~ $ aws ec2 create-security-group --group-name carrd12 --description "my security group" --vpc-id vpc-0e2544332a068204f
{
  "GroupId": "sg-0804485cf3bfa3922",
  "SecurityGroupArn": "arn:aws:ec2:us-east-1:881490118823:security-group/sg-0804485cf3bfa3922"
}
~ $

```

Create rules for security group TCP port 22 and TCP port 80:

```

~ $ aws ec2 authorize-security-group-ingress --group-id sg-0804485cf3bfa3922 --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0e6dd085ae052e82b",
      "GroupId": "sg-0804485cf3bfa3922",
      "GroupOwnerId": "881490118823",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0e6dd085ae052e82b"
    }
  ]
}
~ $

```



```

~ $ aws ec2 authorize-security-group-ingress --group-id sg-0804485cf3bfa3922 --protocol tcp --port 80 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0066c3d02581b93eb",
      "GroupId": "sg-0804485cf3bfa3922",
      "GroupOwnerId": "881490118823",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0066c3d02581b93eb"
    }
  ]
}

```

Verify security group:

```

~ $ aws ec2 describe-security-groups --group-id sg-0804485cf3bfa3922
{
  "SecurityGroups": [
    {
      "GroupId": "sg-0804485cf3bfa3922",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-0e2544332a068204f",
      "SecurityGroupArn": "arn:aws:ec2:us-east-1:881490118823:security-group/sg-0804485cf3bfa3922",
      "OwnerId": "881490118823",
      "GroupName": "carrrd12",
      "Description": "my security group",
      "IpPermissions": [
        {
          "IpProtocol": "tcp",
          "FromPort": 80,
          "ToPort": 80
        }
      ]
    }
  ]
}

```

```

      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "UserIdGroupPairs": [],
      "IpRanges": [
        {
          "CidrIp": "0.0.0.0/0"
        }
      ],
      "Ipv6Ranges": [],
      "PrefixListIds": []
    },
    {
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "UserIdGroupPairs": [],
      "IpRanges": [
        {
          "CidrIp": "0.0.0.0/0"
        }
      ],
      "Ipv6Ranges": [],
      "PrefixListIds": []
    }
  ]
}

```

Create key pair and modify permissions:

```

~ $ aws ec2 create-key-pair --key-name carrd12-key --query 'KeyMaterial' --output text > devenv-key.pem
~ $ ls -l devenv-key.pem
-rw-r--r--. 1 cloudshell-user cloudshell-user 1675 Feb 28 14:50 devenv-key.pem
~ $ chmod 400 devenv-key.pem
~ $

```

Run instance:

```

~ $ aws ec2 run-instances --image-id ami-09e67426f25ce0d7 --count 1 --instance-type t3.micro --key-name carrd12-key --security-group-ids sg-0804485cf3bfa3922 --subnet-id subnet-0a06a451
bcdc4099f
{
  "ReservationId": "r-0a31c5d8f273c3498",
  "OwnerId": "881498118823",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "cd95a87d-d6dd-4417-8729-adf6b64e7829",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "x86",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-02-28T14:55:55-08:00",
            "AttachmentId": "eni-attach-0f4da18437d735c30",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0804485cf3bfa3922",
              "GroupName": "carrd12"
            }
          ]
        }
      ]
    }
  ]
}

```

```

    },
    "Description": "",
    "Groups": [
      {
        "GroupId": "sg-0804485cf3bfa3922",
        "GroupName": "carrd12"
      }
    ],
    "Ipv6Addresses": [],
    "MacAddress": "0e:54:36:ea:49:df",
    "NetworkInterfaceId": "eni-0e288f4f61c8c8a43",
    "OwnerId": "881490118823",
    "PrivateDnsName": "ip-172-31-40-251.ec2.internal",
    "PrivateIpAddress": "172.31.40.251",
    "PrivateIpAddresses": [
      {
        "Primary": true,
        "PrivateDnsName": "ip-172-31-40-251.ec2.internal",
        "PrivateIpAddress": "172.31.40.251"
      }
    ],
    "SourceDestCheck": true,
    "Status": "in-use",
    "SubnetId": "subnet-0aa6a451bcda4099f",
    "VpcId": "vpc-0e2544332a068204f",
    "InterfaceType": "interface",
    "Operator": {
      "Managed": false
    }
  }

```

```

    "Managed": false
  }
},
"RootDeviceName": "/dev/sda1",
"RootDeviceType": "ebs",
"SecurityGroups": [
  {
    "GroupId": "sg-0804485cf3bfa3922",
    "GroupName": "carrd12"
  }
],
"SourceDestCheck": true,
"StateReason": {
  "Code": "pending",
  "Message": "pending"
},
"VirtualizationType": "hvm",
"CpuOptions": {
  "CoreCount": 1,
  "ThreadsPerCore": 2
},
"CapacityReservationSpecification": {
  "CapacityReservationPreference": "open"
},
"MetadataOptions": {
  "State": "pending",
  "HttpTokens": "optional",
  "HttpPutResponseHopLimit": 1,

```

```
"MetadataOptions": {
  "State": "pending",
  "HttpTokens": "optional",
  "HttpPutResponseHopLimit": 1,
  "HttpEndpoint": "enabled",
  "HttpProtocolIpv6": "disabled",
  "InstanceMetadataTags": "disabled"
},
"EnclaveOptions": {
  "Enabled": false
},
"PrivateDnsNameOptions": {
  "HostnameType": "ip-name",
  "EnableResourceNameDnsARecord": false,
  "EnableResourceNameDnsAAAARecord": false
},
"MaintenanceOptions": {
  "AutoRecovery": "default"
},
"CurrentInstanceBootMode": "legacy-bios",
"Operator": {
  "Managed": false
},
"InstanceId": "i-08c7ea551987fbc8f",
"ImageId": "ami-09e67e426f25ce0d7",
"State": {
  "Code": 0,
  "Name": "pending"
},
```

```

    "InstanceId": "i-08c7ea551987fbc8f",
    "ImageId": "ami-09e67e426f25ce0d7",
    "State": {
      "Code": 0,
      "Name": "pending"
    },
    "PrivateDnsName": "ip-172-31-40-251.ec2.internal",
    "PublicDnsName": "",
    "StateTransitionReason": "",
    "KeyName": "carrd12-key",
    "AmiLaunchIndex": 0,
    "ProductCodes": [],
    "InstanceType": "t3.micro",
    "LaunchTime": "2025-02-28T14:55:55+00:00",
    "Placement": {
      "GroupName": "",
      "Tenancy": "default",
      "AvailabilityZone": "us-east-1a"
    },
    "Monitoring": {
      "State": "disabled"
    },
    "SubnetId": "subnet-0aa6a451bcda4099f",
    "VpcId": "vpc-0e2544332a068204f",
    "PrivateIpAddress": "172.31.40.251"
  }
]
}
~ $ █

```

Verify Instance:

```

~ $ aws ec2 describe-instances --instance-ids i-08c7ea551987fbc8f
{
  "Reservations": [
    {
      "ReservationId": "r-0a31c5a8f273c3490",
      "OwnerId": "881490118823",
      "Groups": [],
      "Instances": [
        {
          "Architecture": "x86_64",
          "BlockDeviceMappings": [
            {
              "DeviceName": "/dev/sda1",
              "Ebs": {
                "AttachTime": "2025-02-28T14:55:56+00:00",
                "DeleteOnTermination": true,
                "Status": "attached",
                "VolumeId": "vol-05f84f2224d06044a"
              }
            }
          ],
          "ClientToken": "ce59d07d-d6dd-4417-8729-adf6b64a7829",
          "EbsOptimized": false,
          "EnaSupport": true,
          "Hypervisor": "xen",
          "NetworkInterfaces": [
            {
              "Association": {
                "IpOwnerId": "amazon",

```

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```
"Association": {
  "IpOwnerId": "amazon",
  "PublicDnsName": "ec2-18-215-239-60.compute-1.amazonaws.com",
  "PublicIp": "18.215.239.60"
},
"Attachment": {
  "AttachTime": "2025-02-28T14:55:55+00:00",
  "AttachmentId": "eni-attach-0f4da18437d715c30",
  "DeleteOnTermination": true,
  "DeviceIndex": 0,
  "Status": "attached",
  "NetworkCardIndex": 0
},
>Description": "",
"Groups": [
  {
    "GroupId": "sg-0804485cf3bfa3922",
    "GroupName": "carrd12"
  }
],
"Ipv6Addresses": [],
"MacAddress": "0e:54:36:ea:49:df",
"NetworkInterfaceId": "eni-0e288f4f61c8c8a43",
"OwnerId": "881490118823",
"PrivateDnsName": "ip-172-31-40-251.ec2.internal",
"PrivateIpAddress": "172.31.40.251",
"PrivateIpAddresses": [
  {
    "Association": {
```

AWS CloudShell

```
"Association": {
  "IpOwnerId": "amazon",
  "PublicDnsName": "ec2-18-215-239-60.compute-1.amazonaws.com",
  "PublicIp": "18.215.239.60"
},
"Primary": true,
"PrivateDnsName": "ip-172-31-40-251.ec2.internal",
"PrivateIpAddress": "172.31.40.251"
}
],
"SourceDestCheck": true,
"Status": "in-use",
"SubnetId": "subnet-0aa6a451bcda4099f",
"VpcId": "vpc-0e2544332a068204f",
"InterfaceType": "interface",
"Operator": {
  "Managed": false
}
}
],
"RootDeviceName": "/dev/sda1",
"RootDeviceType": "ebs",
"SecurityGroups": [
  {
    "GroupId": "sg-0804485cf3bfa3922",
    "GroupName": "carrd12"
  }
],
"SourceDestCheck": true,
```

```

"SourceDestCheck": true,
"VirtualizationType": "hvm",
"CpuOptions": {
  "CoreCount": 1,
  "ThreadsPerCore": 2
},
"CapacityReservationSpecification": {
  "CapacityReservationPreference": "open"
},
"HibernationOptions": {
  "Configured": false
},
"MetadataOptions": {
  "State": "applied",
  "HttpTokens": "optional",
  "HttpPutResponseHopLimit": 1,
  "HttpEndpoint": "enabled",
  "HttpProtocolIpv6": "disabled",
  "InstanceMetadataTags": "disabled"
},
"EnclaveOptions": {
  "Enabled": false
},
"PlatformDetails": "Linux/UNIX",
"UsageOperation": "RunInstances",
"UsageOperationUpdateTime": "2025-02-28T14:55:55+00:00",
"PrivateDnsNameOptions": {
  "HostnameType": "ip-name",
  "EnableResourceNameDnsARecord": false,

```

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```

"EnableResourceNameDnsARecord": false,
"EnableResourceNameDnsAAAARecord": false
},
"MaintenanceOptions": {
  "AutoRecovery": "default"
},
"CurrentInstanceBootMode": "legacy-bios",
"NetworkPerformanceOptions": {
  "BandwidthWeighting": "default"
},
"Operator": {
  "Managed": false
},
"InstanceId": "i-08c7ea551987fbc8f",
"ImageId": "ami-09e67e426f25ce0d7",
"State": {
  "Code": 16,
  "Name": "running"
},
"PrivateDnsName": "ip-172-31-40-251.ec2.internal",
"PublicDnsName": "ec2-18-215-239-60.compute-1.amazonaws.com",
"StateTransitionReason": "",
"KeyName": "carrd12-key",
"AmiLaunchIndex": 0,
"ProductCodes": [],
"InstanceType": "t3.micro",
"LaunchTime": "2025-02-28T14:55:55+00:00",
"Placement": {
  "GroupName": "",

```

```

    "Code": 16,
    "Name": "running"
  },
  "PrivateDnsName": "ip-172-31-40-251.ec2.internal",
  "PublicDnsName": "ec2-18-215-239-60.compute-1.amazonaws.com",
  "StateTransitionReason": "",
  "KeyName": "carrd12-key",
  "AmiLaunchIndex": 0,
  "ProductCodes": [],
  "InstanceType": "t3.micro",
  "LaunchTime": "2025-02-28T14:55:55+00:00",
  "Placement": {
    "GroupName": "",
    "Tenancy": "default",
    "AvailabilityZone": "us-east-1a"
  },
  "Monitoring": {
    "State": "disabled"
  },
  "SubnetId": "subnet-0aa6a451bcda4099f",
  "VpcId": "vpc-0e2544332a068204f",
  "PrivateIpAddress": "172.31.40.251",
  "PublicIpAddress": "18.215.239.60"
}
]
}
}
~ $

```

Use the following ssh command to connect to the instance:

```

~ $ ssh -i devenv-key.pem ubuntu@18.215.239.60
The authenticity of host '18.215.239.60 (18.215.239.60)' can't be established.
ED25519 key fingerprint is SHA256:l2Btq32BhYbJwBkxjARSyeH4pefmvKt/cICAstqSig0.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.215.239.60' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-1045-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Fri Feb 28 15:03:59 UTC 2025

System load:  0.0           Processes:    104
Usage of /:   16.4% of 7.69GB Users logged in: 0
Memory usage: 23%          IPv4 address for ens5: 172.31.40.251
Swap usage:   0%

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the

```

Switch to root user

```

ubuntu@ip-172-31-40-251:~$ sudo -s
root@ip-172-31-40-251:/home/ubuntu#

```

Verify Cpu info:


```

root@ip-172-31-40-251:/home/ubuntu# cat /proc/cpuinfo
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 85
model name     : Intel(R) Xeon(R) Platinum 8259C CPU @ 2.50GHz
stepping       : 7
microcode     : 0x5003801
cpu MHz        : 2499.996
cache size     : 36088 KB
physical id    : 0
siblings       : 2
core id        : 0
cpu cores      : 1
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_
good_nopl xtology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq sse3 fma cx16 pcid sse4.1 sse4.2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdand hy
pervisor_lahf_lm dsm 3dnowprefetch invpcid_single pti fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid mpx avx512f avx512dq rdseed adx snap clflushopt clwb avx512cd
avx512bw avx512vb_1 xsaveopt xsavec xgetbv1 xsaves ida arat plru opsie
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapsgs itlb_multihit
bogomips       : 4999.99
clflush size   : 64
cache alignment : 64
address sizes   : 46 bits physical, 48 bits virtual

```

```

processor       : 1
vendor_id      : GenuineIntel
cpu family     : 6
model          : 85
model name     : Intel(R) Xeon(R) Platinum 8259C CPU @ 2.50GHz
stepping       : 7
microcode     : 0x5003801
cpu MHz        : 2499.996
cache size     : 36088 KB
physical id    : 0
siblings       : 2
core id        : 0
cpu cores      : 1
apicid         : 1
initial apicid : 1
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_
good_nopl xtology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq sse3 fma cx16 pcid sse4.1 sse4.2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdand hy
pervisor_lahf_lm dsm 3dnowprefetch invpcid_single pti fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid mpx avx512f avx512dq rdseed adx snap clflushopt clwb avx512cd
avx512bw avx512vb_1 xsaveopt xsavec xgetbv1 xsaves ida arat plru opsie
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapsgs itlb_multihit
bogomips       : 4999.99
clflush size   : 64
cache alignment : 64
address sizes   : 46 bits physical, 48 bits virtual
power management:

```

Run the following commands to install and start an Apache web server:

```

root@ip-172-31-40-251:/home/ubuntu# apt update -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/universe amd64 Packages [8628 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/universe Translation-en [5124 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/universe amd64 c-n-f Metadata [265 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/multiverse amd64 Packages [144 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/multiverse Translation-en [104 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/multiverse amd64 c-n-f Metadata [9136 B]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3799 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main Translation-en [579 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [17.8 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3593 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [503 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [548 B]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1257 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [302 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [28.3 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [27.9 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [7968 B]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [612 B]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45.7 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/main Translation-en [16.3 kB]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1420 B]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/restricted amd64 c-n-f Metadata [116 B]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [25.0 kB]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16.3 kB]

```

```
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [28.3 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [27.9 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [7968 B]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [612 B]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45.7 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/main Translation-en [16.3 kB]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1420 B]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/restricted amd64 c-n-f Metadata [116 B]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [25.0 kB]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16.3 kB]
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:31 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3419 kB]
Get:32 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [499 kB]
Get:33 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [14.3 kB]
Get:34 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3446 kB]
Get:35 http://security.ubuntu.com/ubuntu focal-security/restricted Translation-en [483 kB]
Get:36 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [548 B]
Get:37 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1035 kB]
Get:38 http://security.ubuntu.com/ubuntu focal-security/universe Translation-en [219 kB]
Get:39 http://security.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [21.4 kB]
Get:40 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [24.8 kB]
Get:41 http://security.ubuntu.com/ubuntu focal-security/multiverse Translation-en [5968 B]
Get:42 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [540 B]
Fetched 34.0 MB in 7s (5193 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
277 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
root@ip-172-31-40-251:/home/ubuntu# apt full-upgrade -y
```

```
....
```

```
Unpacking python3-problem-report (2.20.11-0ubuntu27.27) over (2.20.11-0ubuntu27.16) ...
Preparing to unpack .../066-python3-apport_2.20.11-0ubuntu27.27_all.deb ...
Unpacking python3-apport (2.20.11-0ubuntu27.27) over (2.20.11-0ubuntu27.16) ...
Preparing to unpack .../067-apport_2.20.11-0ubuntu27.27_all.deb ...
Unpacking apport (2.20.11-0ubuntu27.27) over (2.20.11-0ubuntu27.16) ...
Preparing to unpack .../068-bolt_0.9.1-2-ubuntu20.04.2_amd64.deb ...
Unpacking bolt (0.9.1-2-ubuntu20.04.2) over (0.8-4ubuntu1) ...
Preparing to unpack .../069-cryptsetup-initramfs_2%3a2.2.2-3ubuntu2.4_all.deb ...
Unpacking cryptsetup-initramfs (2:2.2.2-3ubuntu2.4) over (2:2.2.2-3ubuntu2.3) ...
Preparing to unpack .../070-cryptsetup-bin_2%3a2.2.2-3ubuntu2.4_amd64.deb ...
Unpacking cryptsetup-bin (2:2.2.2-3ubuntu2.4) over (2:2.2.2-3ubuntu2.3) ...
Preparing to unpack .../071-cryptsetup_2%3a2.2.2-3ubuntu2.4_amd64.deb ...
Unpacking cryptsetup (2:2.2.2-3ubuntu2.4) over (2:2.2.2-3ubuntu2.3) ...
Preparing to unpack .../072-cryptsetup-run_2%3a2.2.2-3ubuntu2.4_all.deb ...
Unpacking cryptsetup-run (2:2.2.2-3ubuntu2.4) over (2:2.2.2-3ubuntu2.3) ...
Preparing to unpack .../073-curl_7.68.0-1ubuntu2.25_amd64.deb ...
Unpacking curl (7.68.0-1ubuntu2.25) over (7.68.0-1ubuntu2.5) ...
Preparing to unpack .../074-libcurl4_7.68.0-1ubuntu2.25_amd64.deb ...
Unpacking libcurl4:amd64 (7.68.0-1ubuntu2.25) over (7.68.0-1ubuntu2.5) ...
Preparing to unpack .../075-ec2-hibinit-agent_1.0.0-0ubuntu9.6_all.deb ...
Unpacking ec2-hibinit-agent (1.0.0-0ubuntu9.6) over (1.0.0-0ubuntu9.1) ...
Preparing to unpack .../076-fwupd-signed_1.51.1-20.04.1+1.4-0ubuntu0.1_amd64.deb ...
Unpacking fwupd-signed (1.51.1-20.04.1+1.4-0ubuntu0.1) over (1.27.1ubuntu2+1.3.11-1-focal1) ...
Preparing to unpack .../077-libgpgme11_1.13.1-7ubuntu2.2_amd64.deb ...
Unpacking libgpgme11:amd64 (1.13.1-7ubuntu2.2) over (1.13.1-7ubuntu2) ...
Selecting previously unselected package libjcat1:amd64.
Preparing to unpack .../078-libjcat1_0.1.4-0ubuntu0.20.04.1_amd64.deb ...
Unpacking libjcat1:amd64 (0.1.4-0ubuntu0.20.04.1) ...
Preparing to unpack .../079-libfwupd2_1.7.9-1-20.04.3_amd64.deb ...
Unpacking libfwupd2:amd64 (1.7.9-1-20.04.3) over (1.3.11-1-focal1) ...
Preparing to unpack .../080-libarchive13_3.4.0-2ubuntu1.4_amd64.deb ...
Unpacking libarchive13:amd64 (3.4.0-2ubuntu1.4) over (3.4.0-2ubuntu1) ...
Selecting previously unselected package libxmlb2:amd64.
Preparing to unpack .../081-libxmlb2_0.3.6-2build1-20.04.1_amd64.deb ...
Unpacking libxmlb2:amd64 (0.3.6-2build1-20.04.1) ...
Selecting previously unselected package libfwupdplugin5:amd64.
Preparing to unpack .../082-libfwupdplugin5_1.7.9-1-20.04.3_amd64.deb ...
Unpacking libfwupdplugin5:amd64 (1.7.9-1-20.04.3) ...
Selecting previously unselected package libmbim-glib4:amd64.
Preparing to unpack .../083-libmbim-glib4_1.26.2-1-ubuntu20.04.1_amd64.deb ...
Unpacking libmbim-glib4:amd64 (1.26.2-1-ubuntu20.04.1) ...
Selecting previously unselected package libmbim-proxy.
Preparing to unpack .../084-libmbim-proxy_1.26.2-1-ubuntu20.04.1_amd64.deb ...
Unpacking libmbim-proxy (1.26.2-1-ubuntu20.04.1) ...
Selecting previously unselected package libmm-glib0:amd64.
Preparing to unpack .../085-libmm-glib0_1.18.6-1-ubuntu20.04.1_amd64.deb ...
Unpacking libmm-glib0:amd64 (1.18.6-1-ubuntu20.04.1) ...
Selecting previously unselected package libqmi-glib5:amd64.
Preparing to unpack .../086-libqmi-glib5_1.30.4-1-ubuntu20.04.1_amd64.deb ...
Unpacking libqmi-glib5:amd64 (1.30.4-1-ubuntu20.04.1) ...
Selecting previously unselected package libqmi-proxy.
Preparing to unpack .../087-libqmi-proxy_1.30.4-1-ubuntu20.04.1_amd64.deb ...
Unpacking libqmi-proxy (1.30.4-1-ubuntu20.04.1) ...
Preparing to unpack .../088-libtss2-esys0_2.3.2-1ubuntu0.20.04.2_amd64.deb ...
```

Unpacking libtss2-esys0 (2.3.2-1ubuntu0.20.04.2) over (2.3.2-1) ...
Preparing to unpack .../089-fwupd_1.7.9-1~20.04.3_amd64.deb ...
Unpacking fwupd (1.7.9-1~20.04.3) over (1.3.11-1~focal1) ...
Preparing to unpack .../090-hibagent_1.0.1-0ubuntu1.20.04.2_all.deb ...
Unpacking hibagent (1.0.1-0ubuntu1.20.04.2) over (1.0.1-0ubuntu1) ...
Preparing to unpack .../091-python3-twisted-bin_18.9.0-11ubuntu0.20.04.5_amd64.deb ...
Unpacking python3-twisted-bin:amd64 (18.9.0-11ubuntu0.20.04.5) over (18.9.0-11ubuntu0.20.04.1) ...
Preparing to unpack .../092-python3-idna_2.8-1ubuntu0.1_all.deb ...
Unpacking python3-idna (2.8-1ubuntu0.1) over (2.8-1) ...
Preparing to unpack .../093-python3-twisted_18.9.0-11ubuntu0.20.04.5_all.deb ...
Unpacking python3-twisted (18.9.0-11ubuntu0.20.04.5) over (18.9.0-11ubuntu0.20.04.1) ...
Preparing to unpack .../094-python3-configobj_5.0.6-4ubuntu0.1_all.deb ...
Unpacking python3-configobj (5.0.6-4ubuntu0.1) over (5.0.6-4) ...
Preparing to unpack .../095-landscape-common_23.02-0ubuntu1~20.04.4_amd64.deb ...
Unpacking landscape-common (23.02-0ubuntu1~20.04.4) over (19.12-0ubuntu4.2) ...
Selecting previously unselected package libatasmart4:amd64.
Preparing to unpack .../096-libatasmart4_0.19-5_amd64.deb ...
Unpacking libatasmart4:amd64 (0.19-5) ...
Selecting previously unselected package libblockdev-utils2:amd64.
Preparing to unpack .../097-libblockdev-utils2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev-utils2:amd64 (2.23-2ubuntu3) ...
Selecting previously unselected package libnspr4:amd64.
Preparing to unpack .../098-libspr4_2%3a4.35-0ubuntu0.20.04.1_amd64.deb ...
Unpacking libnspr4:amd64 (2.4.35-0ubuntu0.20.04.1) ...
Selecting previously unselected package libnss3:amd64.
Preparing to unpack .../099-libnss3_2%3a3.98-0ubuntu0.20.04.2_amd64.deb ...
Unpacking libnss3:amd64 (2.3.98-0ubuntu0.20.04.2) ...
Selecting previously unselected package libvolume-key1.
Preparing to unpack .../100-libvolume-key1_0.3.12-3.1_amd64.deb ...
Unpacking libvolume-key1 (0.3.12-3.1) ...
Selecting previously unselected package libblockdev-crypto2:amd64.
Preparing to unpack .../101-libblockdev-crypto2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev-crypto2:amd64 (2.23-2ubuntu3) ...
Selecting previously unselected package libblockdev-part-err2:amd64.
Preparing to unpack .../102-libblockdev-part-err2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev-part-err2:amd64 (2.23-2ubuntu3) ...
Selecting previously unselected package libparted-fs-resize0:amd64.
Preparing to unpack .../103-libparted-fs-resize0_3.3-4ubuntu0.20.04.1_amd64.deb ...
Unpacking libparted-fs-resize0:amd64 (3.3-4ubuntu0.20.04.1) ...
Selecting previously unselected package libblockdev-fs2:amd64.
Preparing to unpack .../104-libblockdev-fs2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev-fs2:amd64 (2.23-2ubuntu3) ...
Selecting previously unselected package libblockdev-loop2:amd64.
Preparing to unpack .../105-libblockdev-loop2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev-loop2:amd64 (2.23-2ubuntu3) ...
Selecting previously unselected package libblockdev-part2:amd64.
Preparing to unpack .../106-libblockdev-part2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev-part2:amd64 (2.23-2ubuntu3) ...
Selecting previously unselected package libblockdev-swap2:amd64.
Preparing to unpack .../107-libblockdev-swap2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev-swap2:amd64 (2.23-2ubuntu3) ...
Selecting previously unselected package libblockdev2:amd64.
Preparing to unpack .../108-libblockdev2_2.23-2ubuntu3_amd64.deb ...
Unpacking libblockdev2:amd64 (2.23-2ubuntu3) ...
Preparing to unpack .../109-libxmlb1_0.1.15-2ubuntu1~20.04.1_amd64.deb ...
Unpacking libxmlb1:amd64 (0.1.15-2ubuntu1~20.04.1) over (0.1.15-2) ...
Preparing to unpack .../110-libfwupdplugin1_1.5.11-0ubuntu1~20.04.2_amd64.deb ...
Unpacking libfwupdplugin1:amd64 (1.5.11-0ubuntu1~20.04.2) over (1.3.11-1~focal1) ...
Preparing to unpack .../111-libgstreamer1.0-0_1.16.3-0ubuntu1.2_amd64.deb ...
Unpacking libgstreamer1.0-0:amd64 (1.16.3-0ubuntu1.2) over (1.16.2-2) ...
Preparing to unpack .../112-libasf2-modules_2.1.27+dfsg-2ubuntu0.1_amd64.deb ...
Unpacking libasf2-modules:amd64 (2.1.27+dfsg-2ubuntu0.1) over (2.1.27+dfsg-2) ...
Preparing to unpack .../113-libsoup2.4-1_2.70.0-1ubuntu0.1_amd64.deb ...
Unpacking libsoup2.4-1:amd64 (2.70.0-1ubuntu0.1) over (2.70.0-1) ...
Preparing to unpack .../114-libtdb1_1.4.5-0ubuntu0.20.04.1_amd64.deb ...
Unpacking libtdb1:amd64 (1.4.5-0ubuntu0.20.04.1) over (1.4.2-3build1) ...
Selecting previously unselected package libudisks2-0:amd64.
Preparing to unpack .../115-libudisks2-0_2.8.4-1ubuntu2_amd64.deb ...
Unpacking libudisks2-0:amd64 (2.8.4-1ubuntu2) ...
Preparing to unpack .../116-libunwind8_1.2.1-9ubuntu0.1_amd64.deb ...
Unpacking libunwind8:amd64 (1.2.1-9ubuntu0.1) over (1.2.1-9build1) ...
Preparing to unpack .../117-libxslt1.1_1.1.34-4ubuntu0.20.04.1_amd64.deb ...
Unpacking libxslt1.1:amd64 (1.1.34-4ubuntu0.20.04.1) over (1.1.34-4) ...
Selecting previously unselected package linux-modules-5.15.0-1077-aws.
Preparing to unpack .../118-linux-modules-5.15.0-1077-aws_5.15.0-1077.84~20.04.1_amd64.deb ...
Unpacking linux-modules-5.15.0-1077-aws (5.15.0-1077.84~20.04.1) ...
Selecting previously unselected package linux-image-5.15.0-1077-aws.
Preparing to unpack .../119-linux-image-5.15.0-1077-aws_5.15.0-1077.84~20.04.1_amd64.deb ...
Unpacking linux-image-5.15.0-1077-aws (5.15.0-1077.84~20.04.1) ...
Preparing to unpack .../120-linux-aws_5.15.0.1077.84~20.04.1_amd64.deb ...
Unpacking linux-aws (5.15.0.1077.84~20.04.1) over (5.4.0.1045.46) ...

Preparing to unpack .../121-linux-image-aws_5.15.0.1077.84~20.04.1_amd64.deb ...
Unpacking linux-image-aws (5.15.0.1077.84~20.04.1) over (5.4.0.1045.46) ...
Selecting previously unselected package linux-aws-5.15-headers-5.15.0-1077.
Preparing to unpack .../122-linux-aws-5.15-headers-5.15.0-1077_5.15.0-1077.84~20.04.1_all.deb ...
Unpacking linux-aws-5.15-headers-5.15.0-1077 (5.15.0-1077.84~20.04.1) ...
Selecting previously unselected package linux-headers-5.15.0-1077-aws.
Preparing to unpack .../123-linux-headers-5.15.0-1077-aws_5.15.0-1077.84~20.04.1_amd64.deb ...
Unpacking linux-headers-5.15.0-1077-aws (5.15.0-1077.84~20.04.1) ...
Preparing to unpack .../124-linux-headers-aws_5.15.0.1077.84~20.04.1_amd64.deb ...
Unpacking linux-headers-aws (5.15.0.1077.84~20.04.1) over (5.4.0.1045.46) ...
Selecting previously unselected package modemmanager.
Preparing to unpack .../125-modemmanager_1.18.6-1~ubuntu20.04.1_amd64.deb ...
Unpacking modemmanager (1.18.6-1~ubuntu20.04.1) ...
Preparing to unpack .../126-ncurses-term_6.2-0ubuntu2.1_all.deb ...
Unpacking ncurses-term (6.2-0ubuntu2.1) over (6.2-0ubuntu2) ...
Preparing to unpack .../127-pollinate_4.33-3ubuntu1.20.04.2_all.deb ...
Unpacking pollinate (4.33-3ubuntu1.20.04.2) over (4.33-3ubuntu1.20.04.1) ...
Preparing to unpack .../128-python3-cryptography_2.8-3ubuntu0.3_amd64.deb ...
Unpacking python3-cryptography (2.8-3ubuntu0.3) over (2.8-3ubuntu0.1) ...
Preparing to unpack .../129-python3-jinja2_2.10.1-2ubuntu0.4_all.deb ...
Unpacking python3-jinja2 (2.10.1-2ubuntu0.4) over (2.10.1-2) ...
Preparing to unpack .../130-python3-jwt_1.7.1-2ubuntu2.1_all.deb ...
Unpacking python3-jwt (1.7.1-2ubuntu2.1) over (1.7.1-2ubuntu2) ...
Selecting previously unselected package python3-pyparsing.
Preparing to unpack .../131-python3-pyparsing_2.4.6-1_all.deb ...
Unpacking python3-pyparsing (2.4.6-1) ...
Selecting previously unselected package python3-packaging.
Preparing to unpack .../132-python3-packaging_20.3-1_all.deb ...
Unpacking python3-packaging (20.3-1) ...
Preparing to unpack .../133-python3-urllib3_1.25.8-2ubuntu0.4_all.deb ...
Unpacking python3-urllib3 (1.25.8-2ubuntu0.4) over (1.25.8-2ubuntu0.1) ...
Preparing to unpack .../134-python3-requests_2.22.0-2ubuntu1.1_all.deb ...
Unpacking python3-requests (2.22.0-2ubuntu1.1) over (2.22.0-2ubuntu1) ...
Preparing to unpack .../135-software-properties-common_0.99.9.12_all.deb ...
Unpacking software-properties-common (0.99.9.12) over (0.98.9.4) ...
Preparing to unpack .../136-python3-software-properties_0.99.9.12_all.deb ...
Unpacking python3-software-properties (0.99.9.12) over (0.98.9.4) ...
Preparing to unpack .../137-python3-zipp_1.0.0-1ubuntu0.1_all.deb ...
Unpacking python3-zipp (1.0.0-1ubuntu0.1) over (1.0.0-1) ...
Preparing to unpack .../138-sbsigntool_0.9.2-2ubuntu1.1_amd64.deb ...
Unpacking sbsigntool (0.9.2-2ubuntu1.1) over (0.9.2-2ubuntu1) ...
Preparing to unpack .../139-squashfs-tools_1%3a4.4-1ubuntu0.3_amd64.deb ...
Unpacking squashfs-tools (1:4.4-1ubuntu0.3) over (1:4.4-1) ...
Preparing to unpack .../140-snapd_2.66.1+20.04_amd64.deb ...
Unpacking snapd (2.66.1+20.04) over (2.48.3+20.04) ...
Preparing to unpack .../141-sosreport_4.7.2-0ubuntu1~20.04.2_amd64.deb ...
Unpacking sosreport (4.7.2-0ubuntu1~20.04.2) over (4.1-1ubuntu0.20.04.1) ...
Preparing to unpack .../142-tmux_3.0a-2ubuntu0.4_amd64.deb ...
Unpacking tmux (3.0a-2ubuntu0.4) over (3.0a-2ubuntu0.2) ...
Selecting previously unselected package udisks2.
Preparing to unpack .../143-udisks2_2.8.4-1ubuntu2_amd64.deb ...
Unpacking udisks2 (2.8.4-1ubuntu2) ...
Preparing to unpack .../144-unattended-upgrades_2.3ubuntu0.3_all.deb ...
Unpacking unattended-upgrades (2.3ubuntu0.3) over (2.3ubuntu0.1) ...
Preparing to unpack .../145-xfsprogs_5.3.0-1ubuntu2.1_amd64.deb ...
Unpacking xfsprogs (5.3.0-1ubuntu2.1) over (5.3.0-1ubuntu2) ...
Preparing to unpack .../146-amd64-microcode_3.20191218.1ubuntu1.3_amd64.deb ...
Unpacking amd64-microcode (3.20191218.1ubuntu1.3) over (3.20191218.1ubuntu1) ...
Preparing to unpack .../147-cloud-init_24.4.1-0ubuntu0~20.04.1_all.deb ...
Unpacking cloud-init (24.4.1-0ubuntu0~20.04.1) over (21.1-19-gbad84ad4-0ubuntu1~20.04.2) ...
dpkg: warning: unable to delete old directory '/etc/NetworkManager/dispatcher.d': Directory not empty
dpkg: warning: unable to delete old directory '/etc/NetworkManager': Directory not empty
Preparing to unpack .../148-cloud-initramfs-copymods_0.45ubuntu2_all.deb ...
Unpacking cloud-initramfs-copymods (0.45ubuntu2) over (0.45ubuntu1) ...
Preparing to unpack .../149-cloud-initramfs-dyn-netconf_0.45ubuntu2_all.deb ...
Unpacking cloud-initramfs-dyn-netconf (0.45ubuntu2) over (0.45ubuntu1) ...
Preparing to unpack .../150-intel-microcode_3.20250211.0ubuntu0.20.04.1_amd64.deb ...
Unpacking intel-microcode (3.20250211.0ubuntu0.20.04.1) over (3.20201110.0ubuntu0.20.04.2) ...
Preparing to unpack .../151-kpartx_0.8.3-1ubuntu2.4_amd64.deb ...
Unpacking kpartx (0.8.3-1ubuntu2.4) over (0.8.3-1ubuntu2) ...
Preparing to unpack .../152-multipath-tools_0.8.3-1ubuntu2.4_amd64.deb ...
Warning: Stopping multipath-tools.service, but it can still be activated by:
multipathd.socket
Unpacking multipath-tools (0.8.3-1ubuntu2.4) over (0.8.3-1ubuntu2) ...
Preparing to unpack .../153-overlayroot_0.45ubuntu2_all.deb ...
Unpacking overlayroot (0.45ubuntu2) over (0.45ubuntu1) ...
Selecting previously unselected package usb-modeswitch-data.
Preparing to unpack .../154-usb-modeswitch-data_20191128-3_all.deb ...
Unpacking usb-modeswitch-data (20191128-3) ...
Selecting previously unselected package usb-modeswitch.
Preparing to unpack .../155-usb-modeswitch_2.5.2+repack0-2ubuntu3_amd64.deb ...

Unpacking usb-modeswitch (2.5.2+repack0-2ubuntu3) ...
Setting up python3-pkg-resources (45.2.0-1ubuntu0.2) ...
Setting up libip4tc2:amd64 (1.8.4-3ubuntu2.1) ...
Setting up libksba8:amd64 (1.3.5-2ubuntu0.20.04.2) ...
Setting up libexpat1:amd64 (2.2.9-1ubuntu0.8) ...
Setting up cpio (2.13+dfsg-2ubuntu0.4) ...
Setting up motd-news-config (11ubuntu5.8) ...
Setting up libkeyutils1:amd64 (1.6-6ubuntu1.1) ...
Setting up libapparmor1:amd64 (2.13.3-7ubuntu5.4) ...
Setting up apt-utils (2.0.10) ...
Setting up linux-base (4.5ubuntu3.7) ...
Setting up perl-modules-5.30 (5.30.0-9ubuntu0.5) ...
Setting up squashfs-tools (1:4.4-1ubuntu0.3) ...
Setting up python3-jwt (1.7.1-2ubuntu2.1) ...
Setting up alsa-ucm-conf (1.2.2-1ubuntu0.13) ...
Setting up gawk (1:5.0.1+dfsg-1ubuntu0.1) ...
Setting up python3-gdbm:amd64 (3.8.10-0ubuntu1~20.04) ...
Setting up python3-problem-report (2.20.11-0ubuntu27.27) ...
Setting up libip6tc2:amd64 (1.8.4-3ubuntu2.1) ...
Setting up libglib2.0-0:amd64 (2.64.6-1-ubuntu20.04.8) ...
Setting up distro-info-data (0.43ubuntu1.17) ...
Setting up libtdb1:amd64 (1.4.5-0ubuntu0.20.04.1) ...
Setting up libssl1.1:amd64 (1.1.1f-1ubuntu2.24) ...
Setting up intel-microcode (3.20250211.0ubuntu0.20.04.1) ...
update-initramfs: deferring update (trigger activated)
intel-microcode: microcode will be updated at next boot
Setting up libxmlb2:amd64 (0.3.6-2build1~20.04.1) ...
Setting up libsqlite3-0:amd64 (3.31.1-4ubuntu0.6) ...
Setting up libsasl2-modules:amd64 (2.1.27+dfsg-2ubuntu0.1) ...
Setting up rsyslog (8.2001.0-1ubuntu1.3) ...
The user 'syslog' is already a member of 'adm'.
The user 'syslog' is already a member of 'tty'.
Skipping profile in /etc/apparmor.d/disable: usr.sbin.rsyslogd
Setting up python3-twisted-bin:amd64 (18.9.0-11ubuntu0.20.04.5) ...
Setting up libnghttp2-14:amd64 (1.40.0-1ubuntu0.3) ...
Setting up less (551-1ubuntu0.3) ...
Setting up isc-dhcp-client (4.4.1-2.1ubuntu5.20.04.5) ...
Installing new version of config file /etc/apparmor.d/sbin.dhclient ...
Warning from /etc/apparmor.d/sbin.dhclient (/etc/apparmor.d/sbin.dhclient line 4): apparmor_parser: File '/etc/apparmor.d/sbin.dhclient' missing feature abi, falling back to default policy feature abi
Setting up libnetplan0:amd64 (0.104-0ubuntu2~20.04.6) ...
Setting up fwupd-signed (1.51.1~20.04.1+1.4-0ubuntu0.1) ...
Setting up python3-yaml (5.3.1-1ubuntu0.1) ...
Setting up libntfs-3g883 (1:2017.3.23AR.3-3ubuntu1.3) ...
Setting up krb5-locales (1.17-6ubuntu4.8) ...
Setting up python3-zipp (1.0.0-1ubuntu0.1) ...
Setting up hibagent (1.0.1-0ubuntu1.20.04.2) ...
Setting up amd64-microcode (3.20191218.1ubuntu1.3) ...
update-initramfs: deferring update (trigger activated)
amd64-microcode: microcode will be updated at next boot
Setting up locales (2.31-0ubuntu9.17) ...
Generating locales (this might take a while)...
en_US.UTF-8... done
Generation complete.
Setting up libldap-common (2.4.49+dfsg-2ubuntu1.10) ...
Setting up libunwind8:amd64 (1.2.1-9ubuntu0.1) ...
Setting up xxd (2:8.1.2269-1ubuntu5.31) ...
Setting up ntfs-3g (1:2017.3.23AR.3-3ubuntu1.3) ...
Setting up libcap2:amd64 (1:2.32-1ubuntu0.2) ...
Setting up libkrb5support0:amd64 (1.17-6ubuntu4.8) ...
Setting up libsasl2-modules-db:amd64 (2.1.27+dfsg-2ubuntu0.1) ...
Setting up tzdata (2024b-0ubuntu0.20.04.1) ...

Current default time zone: 'Etc/UTC'

Local time is now: Fri Feb 28 15:16:04 UTC 2025.

Universal Time is now: Fri Feb 28 15:16:04 UTC 2025.

Run 'dpkg-reconfigure tzdata' if you wish to change it.

Setting up libcap2-bin (1:2.32-1ubuntu0.2) ...

Setting up libasound2-data (1.2.2-2.1ubuntu2.5) ...

Setting up apparmor (2.13.3-7ubuntu5.4) ...

Installing new version of config file /etc/apparmor/parser.conf ...

Reloading AppArmor profiles

Skipping profile in /etc/apparmor.d/disable: usr.sbin.rsyslogd

Setting up libklibc:amd64 (2.0.7-1ubuntu5.2) ...

Setting up python3-jinja2 (2.10.1-2ubuntu0.4) ...

Setting up libglib2.0-data (2.64.6-1-ubuntu20.04.8) ...

Setting up libuv1:amd64 (1.34.2-1ubuntu1.5) ...

Setting up python3-configobj (5.0.6-4ubuntu0.1) ...

Setting up vim-common (2:8.1.2269-1ubuntu5.31) ...

Setting up libatasmart4:amd64 (0.19-5) ...

Setting up python3-pyparsing (2.4.6-1) ...
Setting up busybox-static (1:1.30.1-4ubuntu6.5) ...
Setting up libfreetype6:amd64 (2.10.1-2ubuntu0.3) ...
Setting up libpcap0.8:amd64 (1.9.1-3ubuntu1.20.04.1) ...
Setting up libx11-data (2:1.6.9-2ubuntu1.6) ...
Setting up libnspr4:amd64 (2:4.35-0ubuntu0.20.04.1) ...
Setting up gnupg-110n (2.2.19-3ubuntu2.2) ...
Setting up python3-cryptography (2.8-3ubuntu0.3) ...
Setting up libncurses6:amd64 (6.2-0ubuntu2.1) ...
Setting up python3-debian (0.1.36ubuntu1.1) ...
Setting up libdbus-1-3:amd64 (1.12.16-2ubuntu2.3) ...
Setting up dbus (1.12.16-2ubuntu2.3) ...
A reboot is required to replace the running dbus-daemon.
Please reboot the system when convenient.
Setting up xz-utils (5.2.4-1ubuntu1.1) ...
Setting up libfribidi0:amd64 (1.0.8-2ubuntu0.1) ...
Setting up usb-modeswitch-data (20191128-3) ...
Setting up python3-idna (2.8-1ubuntu0.1) ...
Setting up libss2:amd64 (1.45.5-2ubuntu1.2) ...
Setting up sudo (1.8.31-1ubuntu1.5) ...
Setting up libncursesw6:amd64 (6.2-0ubuntu2.1) ...
Setting up libk5crypto3:amd64 (1.17-6ubuntu4.8) ...
Setting up busybox-initramfs (1:1.30.1-4ubuntu6.5) ...
Setting up libxtables12:amd64 (1.8.4-3ubuntu2.1) ...
Setting up logsave (1.45.5-2ubuntu1.2) ...
Setting up libsass2-2:amd64 (2.1.27+dfsg-2ubuntu0.1) ...
Setting up libroken18-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up lz4 (1.9.2-2ubuntu0.20.04.1) ...
Setting up python3-urllib3 (1.25.8-2ubuntu0.4) ...
Setting up libfdisk1:amd64 (2.34-0.1ubuntu9.6) ...
Setting up nano (4.8-1ubuntu1.1) ...
Setting up python-apt-common (2.0.1ubuntu0.20.04.1) ...
Setting up libparted-fs-resize0:amd64 (3.3-4ubuntu0.20.04.1) ...
Setting up mount (2.34-0.1ubuntu9.6) ...
Setting up uuid-runtime (2.34-0.1ubuntu9.6) ...
Setting up libmm-glib0:amd64 (1.18.6-1~ubuntu20.04.1) ...
Setting up libasound2:amd64 (1.2.2-2.1ubuntu2.5) ...
Setting up gpgconf (2.2.19-3ubuntu2.2) ...
Setting up libblockdev-part-err2:amd64 (2.23-2ubuntu3) ...
Setting up libperl5.30:amd64 (5.30.0-9ubuntu0.5) ...
Setting up git-man (1:2.25.1-1ubuntu3.14) ...
Setting up libx11-6:amd64 (2:1.6.9-2ubuntu1.6) ...
Setting up libcryptsetup12:amd64 (2:2.2.2-3ubuntu2.4) ...
Setting up isc-dhcp-common (4.4.1-2.1ubuntu5.20.04.5) ...
Setting up libkrb5-3:amd64 (1.17-6ubuntu4.8) ...
Setting up usb-modeswitch (2.5.2+repack0-2ubuntu3) ...
Setting up sbsigntool (0.9.2-2ubuntu1.1) ...
Setting up libaccountsservice0:amd64 (0.6.55-0ubuntu12~20.04.7) ...
Setting up vim-runtime (2:8.1.2269-1ubuntu5.31) ...
Setting up klibc-utils (2.0.7-1ubuntu5.2) ...
Setting up openssl (1.1.1f-1ubuntu2.24) ...
Setting up python3-lib2to3 (3.8.10-0ubuntu1~20.04) ...
Setting up libdrm-common (2.4.107-8ubuntu1~20.04.2) ...
Setting up libelf1:amd64 (0.176-1.1ubuntu0.1) ...
Setting up libpam-cap:amd64 (1:2.32-1ubuntu0.2) ...
Setting up python3-distro-info (0.23ubuntu1.1) ...
Setting up iputils-ping (3:20190709-3ubuntu1) ...
Setting up tmux (3.0a-2ubuntu0.4) ...
Setting up gpg (2.2.19-3ubuntu2.2) ...
Setting up libpolkit-gobject-1-0:amd64 (0.105-26ubuntu1.3) ...
Setting up libmbim-glib4:amd64 (1.26.2-1~ubuntu20.04.1) ...
Setting up libprocps8:amd64 (2:3.3.16-1ubuntu2.4) ...
Setting up iputils-tracepath (3:20190709-3ubuntu1) ...
Setting up libgstreamer1.0-0:amd64 (1.16.3-0ubuntu1.2) ...
Setcap worked! gst-ptp-helper is not suid!
Setting up rsync (3.1.3-8ubuntu0.9) ...
Setting up libudisks2-0:amd64 (2.8.4-1ubuntu2) ...
Setting up libkmod2:amd64 (27-1ubuntu2.1) ...
Setting up bolt (0.9.1-2~ubuntu20.04.2) ...
bolt.service is a disabled or a static unit not running, not starting it.
Setting up libheimbase1-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up linux-aws-5.15-headers-5.15.0-1077 (5.15.0-1077.84~20.04.1) ...
Setting up gnupg-utils (2.2.19-3ubuntu2.2) ...
Setting up initramfs-tools-bin (0.136ubuntu6.7) ...
Setting up ncurses-term (6.2-0ubuntu2.1) ...
Setting up cryptsetup-bin (2:2.2.2-3ubuntu2.4) ...
Setting up python3-distutils (3.8.10-0ubuntu1~20.04) ...
Setting up networkd-dispatcher (2.1-2~ubuntu20.04.3) ...
Setting up distro-info (0.23ubuntu1.1) ...
Setting up libxmlb1:amd64 (0.1.15-2ubuntu1~20.04.1) ...
Setting up tcpdump (4.9.3-4ubuntu0.3) ...

Installing new version of config file /etc/apparmor.d/usr.sbin.tcpdump ...
Setting up libpython3.8-minimal:amd64 (3.8.10-0ubuntu1~20.04.15) ...
Setting up gpg-agent (2.2.19-3ubuntu2.2) ...
Setting up python3-setuptools (45.2.0-1ubuntu0.2) ...
Setting up wget (1.20.3-1ubuntu2.1) ...
Setting up cryptsetup (2:2.2.2-3ubuntu2.4) ...
Setting up libtss2-esys0 (2.3.2-1ubuntu0.20.04.2) ...
Setting up iptables (1.8.4-3ubuntu2.1) ...
Setting up python3-apt (2.0.1ubuntu0.20.04.1) ...
Setting up libicu66:amd64 (66.1-2ubuntu2.1) ...
Setting up libasn1-8-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up gpgsm (2.2.19-3ubuntu2.2) ...
Setting up ltrace (0.7.3-6.1ubuntu1.1) ...
Setting up libglib2.0-bin (2.64.6-1~ubuntu20.04.8) ...
Setting up e2fsprogs (1.45.5-2ubuntu1.2) ...
update-initramfs: deferring update (trigger activated)
e2scrub_all.service is a disabled or a static unit not running, not starting it.
Setting up ufw (0.36-6ubuntu1.1) ...
Setting up libnss3:amd64 (2:3.98-0ubuntu0.20.04.2) ...
Setting up libmbim-proxy (1.26.2-1~ubuntu20.04.1) ...
Setting up vim-tiny (2:8.1.2269-1ubuntu5.31) ...
Setting up kmod (27-1ubuntu2.1) ...
update-initramfs: deferring update (trigger activated)
Setting up libpgpme11:amd64 (1.13.1-7ubuntu2.2) ...
Setting up libhcrypto4-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up libjcat1:amd64 (0.1.4-0ubuntu0.20.04.1) ...
Setting up fdisk (2.34-0.1ubuntu9.6) ...
Setting up python3-apport (2.20.11-0ubuntu27.27) ...
Setting up grub-common (2.04-1ubuntu26.17) ...
Installing new version of config file /etc/grub.d/10_linux ...
Installing new version of config file /etc/grub.d/20_linux_xen ...
Created symlink /etc/systemd/system/multi-user.target.wants/grub-common.service → /lib/systemd/system/grub-common.service.
Created symlink /etc/systemd/system/sleep.target.wants/grub-common.service → /lib/systemd/system/grub-common.service.
update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults
Setting up ca-certificates (20240203~20.04.1) ...
Updating certificates in /etc/ssl/certs...
rehash: warning: skipping ca-certificates.crt, it does not contain exactly one certificate or CRL
37 added, 20 removed; done.
Setting up unattended-upgrades (2.3ubuntu0.3) ...
Installing new version of config file /etc/kernel/postinst.d/unattended-upgrades ...
Setting up perl (5.30.0-9ubuntu0.5) ...
Setting up libwind0-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up python3-packaging (20.3-1) ...
Setting up cryptsetup-run (2:2.2.2-3ubuntu2.4) ...
Setting up python3-software-properties (0.99.9.12) ...
Setting up libblockdev-utils2:amd64 (2.23-2ubuntu3) ...
Setting up python3-twisted (18.9.0-11ubuntu0.20.04.5) ...
Setting up python3-requests (2.22.0-2ubuntu1.1) ...
Setting up python3-commandnotfound (20.04.6) ...
Setting up linux-headers-5.15.0-1077-aws (5.15.0-1077.84~20.04.1) ...
Setting up libgssapi-krb5-2:amd64 (1.17-6ubuntu4.8) ...
Setting up udev (245.4-4ubuntu3.24) ...
update-initramfs: deferring update (trigger activated)
Setting up xfsprogs (5.3.0-1ubuntu2.1) ...
update-initramfs: deferring update (trigger activated)
Setting up libblockdev-fs2:amd64 (2.23-2ubuntu3) ...
Setting up libssh-4:amd64 (0.9.3-2ubuntu2.5) ...
Setting up sosreport (4.7.2-0ubuntu1~20.04.2) ...
Installing new version of config file /etc/sos/sos.conf ...
Setting up gpg-wks-server (2.2.19-3ubuntu2.2) ...
Setting up libdrm2:amd64 (2.4.107-8ubuntu1~20.04.2) ...
Setting up python3.8-minimal (3.8.10-0ubuntu1~20.04.15) ...
Setting up kpartx (0.8.3-1ubuntu2.4) ...
Setting up procs (2:3.3.16-1ubuntu2.4) ...
Setting up libpolkit-agent-1-0:amd64 (0.105-26ubuntu1.3) ...
Setting up libpython3.8-stdlib:amd64 (3.8.10-0ubuntu1~20.04.15) ...
Setting up initramfs-tools-core (0.136ubuntu6.7) ...
Setting up python3.8 (3.8.10-0ubuntu1~20.04.15) ...
Setting up libxml2:amd64 (2.9.10+dfsg-5ubuntu0.20.04.9) ...
Setting up ubuntu-pro-client (34~20.04) ...
Created symlink /run/cloud-init/cloud-id->/run/cloud-init/cloud-id-aws.
Created symlink /etc/systemd/system/multi-user.target.wants/ua-reboot-cmds.service → /lib/systemd/system/ua-reboot-cmds.service.
Created symlink /etc/systemd/system/timers.target.wants/ua-timer.timer → /lib/systemd/system/ua-timer.timer.
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-advantage.service → /lib/systemd/system/ubuntu-advantage.service.
Setting up accountsservice (0.6.55-0ubuntu12~20.04.7) ...
Setting up ubuntu-pro-client-110n (34~20.04) ...
Setting up libsoup2.4-1:amd64 (2.70.0-1ubuntu0.1) ...
Setting up initramfs-tools (0.136ubuntu6.7) ...
update-initramfs: deferring update (trigger activated)
Setting up libblockdev-swap2:amd64 (2.23-2ubuntu3) ...
Setting up bind9-lb:amd64 (1:9.18.30-0ubuntu0.20.04.2) ...

Setting up libarchive13:amd64 (3.4.0-2ubuntu1.4) ...
Setting up grub2-common (2.04-1ubuntu26.17) ...
Setting up openssh-client (1:8.2p1-4ubuntu0.12) ...
Setting up libhx509-5-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up linux-headers-aws (5.15.0.1077.84-20.04.1) ...
Setting up libblockdev-loop2:amd64 (2.23-2ubuntu3) ...
Setting up libqmi-glib5:amd64 (1.30.4-1-ubuntu20.04.1) ...
Setting up libblockdev2:amd64 (2.23-2ubuntu3) ...
Setting up libblockdev-part2:amd64 (2.23-2ubuntu3) ...
Setting up landscape-common (23.02-0ubuntu1-20.04.4) ...
Setting up grub-pc-bin (2.04-1ubuntu26.17) ...
Setting up open-iscsi (2.0.874-7.1ubuntu6.5) ...
grep: /lib/modules-load.d/open-iscsi.conf: No such file or directory
Setting up software-properties-common (0.99.9.12) ...
Setting up apport (2.20.11-0ubuntu27.27) ...
Installing new version of config file /etc/init.d/apport ...
apport-autoreport.service is a disabled or a static unit, not starting it.
Setting up multipath-tools (0.8.3-1ubuntu2.4) ...
Removing obsolete conffile /etc/init.d/multipath-tools ...
Setting up grub-pc (2.04-1ubuntu26.17) ...
Sourcing file `/etc/default/grub'
Sourcing file `/etc/default/grub.d/40-force-partuuid.cfg'
Sourcing file `/etc/default/grub.d/50-cloudimg-settings.cfg'
Sourcing file `/etc/default/grub.d/init-select.cfg'
Generating grub configuration file ...
GRUB_FORCE_PARTUUID is set, will attempt initrdless boot
Found linux image: /boot/vmlinuz-5.15.0-1077-aws
Found initrd image: /boot/microcode.cpio
Found linux image: /boot/vmlinuz-5.4.0-1045-aws
Found initrd image: /boot/microcode.cpio /boot/initrd.img-5.4.0-1045-aws
Found Ubuntu 20.04.6 LTS (20.04) on /dev/nvme0n1p1
done
Setting up ec2-hibinit-agent (1.0.0-0ubuntu9.6) ...
Installing new version of config file /etc/acpi/actions/sleep.sh ...
Installing new version of config file /etc/hibinit-config.cfg ...
Setting up command-not-found (20.04.6) ...
Setting up libpython3.8:amd64 (3.8.10-0ubuntu1-20.04.15) ...
Setting up cloud-initramfs-copymods (0.45ubuntu2) ...
Setting up open-vm-tools (2:11.3.0-2ubuntu0~ubuntu20.04.7) ...
Installing new version of config file /etc/vmware-tools/tools.conf.example ...
Installing new version of config file /etc/vmware-tools/vgauth.conf ...
Removing obsolete conffile /etc/vmware-tools/vm-support ...
Setting up libxslt1.1:amd64 (1.1.34-4ubuntu0.20.04.1) ...
Setting up cloud-initramfs-dyn-netconf (0.45ubuntu2) ...
Setting up ubuntu-advantage-tools (34-20.04) ...
Setting up bind9-host (1:9.18.30-0ubuntu0.20.04.2) ...
Setting up overlayroot (0.45ubuntu2) ...
Setting up cryptsetup-initramfs (2:2.2.2-3ubuntu2.4) ...
update-initramfs: deferring update (trigger activated)
update-initramfs: deferring update (trigger activated)
Setting up libkrb5-26-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up openssh-sftp-server (1:8.2p1-4ubuntu0.12) ...
Setting up vim (2:8.1.2269-1ubuntu5.31) ...
Setting up openssh-server (1:8.2p1-4ubuntu0.12) ...
rescue-ssh.target is a disabled or a static unit, not starting it.
Setting up libqmi-proxy (1.30.4-1-ubuntu20.04.1) ...
Setting up libheimntlm0-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up libgssapi3-heimdal:amd64 (7.7.0+dfsg-1ubuntu1.4) ...
Setting up bind9-dnsutils (1:9.18.30-0ubuntu0.20.04.2) ...
Setting up libldap-2.4-2:amd64 (2.4.49+dfsg-2ubuntu1.10) ...
Setting up libcurl3-gnutls:amd64 (7.68.0-1ubuntu2.25) ...
Setting up dirmngr (2.2.19-3ubuntu2.2) ...
Setting up git (1:2.25.1-1ubuntu3.14) ...
Setting up libcurl4:amd64 (7.68.0-1ubuntu2.25) ...
Setting up curl (7.68.0-1ubuntu2.25) ...
Setting up gpg-wks-client (2.2.19-3ubuntu2.2) ...
Setting up libfwupd2:amd64 (1.7.9-1-20.04.3) ...
Setting up pollinate (4.33-3ubuntu1.20.04.2) ...
Setting up libfwupdplugin1:amd64 (1.5.11-0ubuntu1-20.04.2) ...
Setting up gnupg (2.2.19-3ubuntu2.2) ...
Setting up libvolume-key1 (0.3.12-3.1) ...
Setting up libfwupdplugin5:amd64 (1.7.9-1-20.04.3) ...
Setting up fwupd (1.7.9-1-20.04.3) ...
Installing new version of config file /etc/fwupd/daemon.conf ...
Installing new version of config file /etc/fwupd/redfish.conf ...
Installing new version of config file /etc/fwupd/remotes.d/lvfs-testing.conf ...
Installing new version of config file /etc/fwupd/remotes.d/lvfs.conf ...
Installing new version of config file /etc/fwupd/thunderbolt.conf ...
Installing new version of config file /etc/update-motd.d/85-fwupd ...
fwupd-offline-update.service is a disabled or a static unit not running, not starting it.
fwupd-refresh.service is a disabled or a static unit not running, not starting it.

fwupd.service is a disabled or a static unit not running, not starting it.
Setting up libblockdev-crypto2:amd64 (2.23-2ubuntu3) ...
Setting up systemd (245.4-4ubuntu3.24) ...
Installing new version of config file /etc/dhcp/dhclient-enter-hooks.d/resolved ...
Setting up python3-distupgrade (1:20.04.41) ...
Setting up netplan.io (0.104-0ubuntu2~20.04.6) ...
Setting up systemd-timesyncd (245.4-4ubuntu3.24) ...
Setting up ubuntu-release-upgrader-core (1:20.04.41) ...
Setting up python3-update-manager (1:20.04.10.23) ...
Setting up linux-image-5.15.0-1077-aws (5.15.0-1077.84~20.04.1) ...
I: /boot/vmlinuz is now a symlink to vmlinuz-5.15.0-1077-aws
I: /boot/initrd.img is now a symlink to initrd.img-5.15.0-1077-aws
Setting up linux-image-aws (5.15.0.1077.84~20.04.1) ...
Setting up linux-modules-5.15.0-1077-aws (5.15.0-1077.84~20.04.1) ...
Setting up systemd-sysv (245.4-4ubuntu3.24) ...
Setting up cloud-init (24.4.1-0ubuntu0~20.04.1) ...
Installing new version of config file /etc/cloud/cloud.cfg ...
Installing new version of config file /etc/cloud/cloud.cfg.d/05_logging.cfg ...
Installing new version of config file /etc/cloud/templates/chef_client.rb.tpl ...
Installing new version of config file /etc/cloud/templates/chrony.conf.alpine.tpl ...
Installing new version of config file /etc/cloud/templates/chrony.conf.debian.tpl ...
Installing new version of config file /etc/cloud/templates/chrony.conf.fedora.tpl ...
Installing new version of config file /etc/cloud/templates/chrony.conf.opensuse.tpl ...
Installing new version of config file /etc/cloud/templates/chrony.conf.rhel.tpl ...
Installing new version of config file /etc/cloud/templates/chrony.conf.sles.tpl ...
Installing new version of config file /etc/cloud/templates/chrony.conf.ubuntu.tpl ...
Installing new version of config file /etc/cloud/templates/hosts.alpine.tpl ...
Installing new version of config file /etc/cloud/templates/hosts.debian.tpl ...
Installing new version of config file /etc/cloud/templates/ntp.conf.alpine.tpl ...
Installing new version of config file /etc/cloud/templates/ntp.conf.debian.tpl ...
Installing new version of config file /etc/cloud/templates/ntp.conf.fedora.tpl ...
Installing new version of config file /etc/cloud/templates/ntp.conf.opensuse.tpl ...
Installing new version of config file /etc/cloud/templates/ntp.conf.rhel.tpl ...
Installing new version of config file /etc/cloud/templates/ntp.conf.sles.tpl ...
Installing new version of config file /etc/cloud/templates/ntp.conf.ubuntu.tpl ...
Installing new version of config file /etc/cloud/templates/resolv.conf.tpl ...
Installing new version of config file /etc/cloud/templates/sources.list.debian.tpl ...
Installing new version of config file /etc/profile.d/Z99-cloud-locale-test.sh ...
Installing new version of config file /etc/profile.d/Z99-cloudinit-warnings.sh ...
Created symlink /etc/systemd/system/cloud-config.target.wants/cloud-init-hotplugd.socket → /lib/systemd/system/cloud-init-hotplugd.socket.
Removing obsolete conffile /etc/NetworkManager/dispatcher.d/hook-network-manager ...
Setting up linux-aws (5.15.0.1077.84~20.04.1) ...
Setting up libnss-systemd:amd64 (245.4-4ubuntu3.24) ...
Setting up update-manager-core (1:20.04.10.23) ...
Setting up libpam-systemd:amd64 (245.4-4ubuntu3.24) ...
Setting up policykit-1 (0.105-26ubuntu1.3) ...
Setting up udisks2 (2.8.4-1ubuntu2) ...
Created symlink /etc/systemd/system/graphical.target.wants/udisks2.service → /lib/systemd/system/udisks2.service.
Setting up update-notifier-common (3.192.30.19) ...
Installing new version of config file /etc/cron.weekly/update-notifier-common ...
Setting up dbus-user-session (1.12.16-2ubuntu2.3) ...
Setting up modemmanager (1.18.6-1~ubuntu20.04.1) ...
Created symlink /etc/systemd/system/dbus-org.freedesktop.ModemManager1.service → /lib/systemd/system/ModemManager.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ModemManager.service → /lib/systemd/system/ModemManager.service.
Setting up snapd (2.66.1+20.04) ...
Installing new version of config file /etc/apparmor.d/usr.lib.snapd.snap-confine.real ...
Installing new version of config file /etc/apt/apt.conf.d/20snapd.conf ...
Installing new version of config file /etc/profile.d/apps-bin-path.sh ...
snapd.failure.service is a disabled or a static unit not running, not starting it.
snapd.snap-repair.service is a disabled or a static unit not running, not starting it.
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.17) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for plymouth-theme-ubuntu-text (0.9.4git20200323-0ubuntu6.2) ...
update-initramfs: deferring update (trigger activated)
Processing triggers for microcode-initrd (2~20.04.0) ...
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
Processing triggers for ca-certificates (20240203~20.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for initramfs-tools (0.136ubuntu6.7) ...
update-initramfs: Generating /boot/initrd.img-5.4.0-1045-aws
Processing triggers for linux-image-5.15.0-1077-aws (5.15.0-1077.84~20.04.1) ...
/etc/kernel/postinst.d/initramfs-tools:
update-initramfs: Generating /boot/initrd.img-5.15.0-1077-aws
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file `/etc/default/grub'
Sourcing file `/etc/default/grub.d/40-force-partuuid.cfg'
Sourcing file `/etc/default/grub.d/50-cloudimg-settings.cfg'

Sourcing file `/etc/default/grub.d/init-select.cfg`

Generating grub configuration file ...

GRUB_FORCE_PARTUUID is set, will attempt initrdless boot

Found linux image: /boot/vmlinuz-5.15.0-1077-aws

Found initrd image: /boot/microcode.cpio /boot/initrd.img-5.15.0-1077-aws

Found linux image: /boot/vmlinuz-5.4.0-1045-aws

Found initrd image: /boot/microcode.cpio /boot/initrd.img-5.4.0-1045-aws

Found Ubuntu 20.04.6 LTS (20.04) on /dev/nvme0n1p1

done

Processing triggers for dbus (1.12.16-2ubuntu2.3) ...

```
root@ip-172-31-40-251:/home/ubuntu# apt install apache2 -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfupdpkgplugin libxslt1
Use 'apt autoremove' to remove them.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser openssl-blacklist
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0 ssl-cert
0 upgraded, 11 newly installed, 0 to remove and 0 not upgraded.
Need to get 1875 kB of archives.
After this operation, 8121 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 libapr1 amd64 1.6.5-1ubuntu1.1 [91.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 libaprutil1 amd64 1.6.1-4ubuntu2.2 [85.1 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-4ubuntu2.2 [10.5 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 libaprutil1-ldap amd64 1.6.1-4ubuntu2.2 [8752 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/main amd64 libjansson4 amd64 2.12-1build1 [28.9 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/main amd64 liblua5.2-0 amd64 5.2.4-1.1build3 [106 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2-bin amd64 2.4.41-4ubuntu3.21 [1189 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2-data all 2.4.41-4ubuntu3.21 [159 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2-utils amd64 2.4.41-4ubuntu3.21 [84.7 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2 amd64 2.4.41-4ubuntu3.21 [95.6 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/main amd64 ssl-cert all 1.0.39 [17.0 kB]
Fetched 1875 kB in 8s (28.8 MB/s)
Preconfiguring packages ...
```

Preconfiguring packages ...

Selecting previously unselected package libapr1:amd64.

(Reading database ... 89684 files and directories currently installed.)

Preparing to unpack .../00-libapr1_1.6.5-1ubuntu1.1_amd64.deb ...

Unpacking libapr1:amd64 (1.6.5-1ubuntu1.1) ...

Selecting previously unselected package libaprutil1:amd64.

Preparing to unpack .../01-libaprutil1_1.6.1-4ubuntu2.2_amd64.deb ...

Unpacking libaprutil1:amd64 (1.6.1-4ubuntu2.2) ...

Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.

Preparing to unpack .../02-libaprutil1-dbd-sqlite3_1.6.1-4ubuntu2.2_amd64.deb ...

Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.1-4ubuntu2.2) ...

Selecting previously unselected package libaprutil1-ldap:amd64.

Preparing to unpack .../03-libaprutil1-ldap_1.6.1-4ubuntu2.2_amd64.deb ...

Unpacking libaprutil1-ldap:amd64 (1.6.1-4ubuntu2.2) ...

Selecting previously unselected package libjansson4:amd64.

Preparing to unpack .../04-libjansson4_2.12-1build1_amd64.deb ...

Unpacking libjansson4:amd64 (2.12-1build1) ...

Selecting previously unselected package liblua5.2-0:amd64.

Preparing to unpack .../05-liblua5.2-0_5.2.4-1.1build3_amd64.deb ...

Unpacking liblua5.2-0:amd64 (5.2.4-1.1build3) ...

Selecting previously unselected package apache2-bin.

Preparing to unpack .../06-apache2-bin_2.4.41-4ubuntu3.21_amd64.deb ...

Unpacking apache2-bin (2.4.41-4ubuntu3.21) ...

Selecting previously unselected package apache2-data.

Preparing to unpack .../07-apache2-data_2.4.41-4ubuntu3.21_all.deb ...

Unpacking apache2-data (2.4.41-4ubuntu3.21) ...

Selecting previously unselected package apache2-utils.

Preparing to unpack .../08-apache2-utils_2.4.41-4ubuntu3.21_amd64.deb ...

Unpacking apache2-utils (2.4.41-4ubuntu3.21) ...

```

Unpacking apache2-utils (2.4.41-4ubuntu3.21) ...
Selecting previously unselected package apache2.
Preparing to unpack .../09-apache2_2.4.41-4ubuntu3.21_amd64.deb ...
Unpacking apache2 (2.4.41-4ubuntu3.21) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../10-ssl-cert_1.0.39_all.deb ...
Unpacking ssl-cert (1.0.39) ...
Setting up libapr1:amd64 (1.6.5-1ubuntu1.1) ...
Setting up libjansson4:amd64 (2.12-1build1) ...
Setting up ssl-cert (1.0.39) ...
Setting up liblua5.2-0:amd64 (5.2.4-1.1build3) ...
Setting up apache2-data (2.4.41-4ubuntu3.21) ...
Setting up libaprutil1:amd64 (1.6.1-4ubuntu2.2) ...
Setting up libaprutil1-ldap:amd64 (1.6.1-4ubuntu2.2) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.1-4ubuntu2.2) ...
Setting up apache2-utils (2.4.41-4ubuntu3.21) ...
Setting up apache2-bin (2.4.41-4ubuntu3.21) ...
Setting up apache2 (2.4.41-4ubuntu3.21) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.

```

```

Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36-6ubuntu1.1) ...
Processing triggers for systemd (245.4-4ubuntu3.24) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.17) ...
root@ip-172-31-40-251:/home/ubuntu#

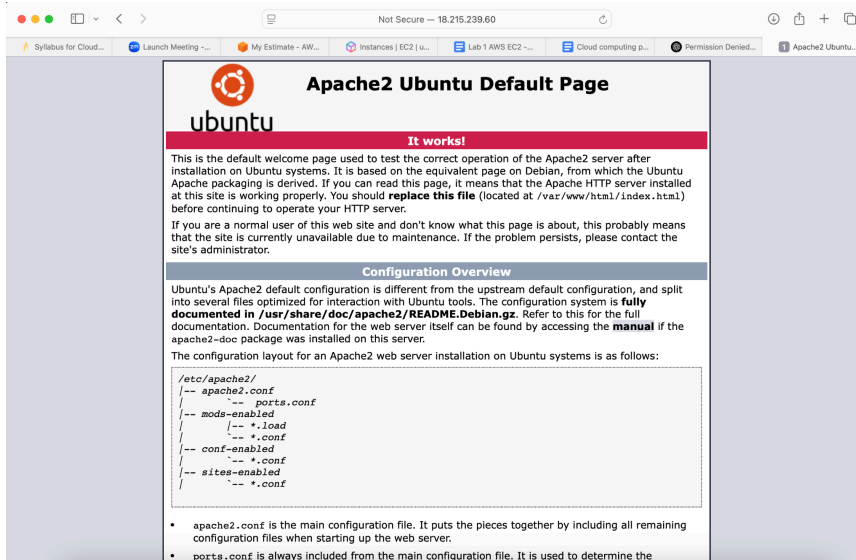
```

```

root@ip-172-31-40-251:/home/ubuntu# /etc/init.d/apache2 restart
Restarting apache2 (via systemctl): apache2.service.
root@ip-172-31-40-251:/home/ubuntu#

```

Access the Apache web server by typing <http://18.215.239.60/> into your browser:



Vi index.html to write the web file

600dd to delete the lines

Press 'i' for insert then enter the desired text

```
<html><body><h1>It works!</h1>
```

```
<p>This website is hosted by Amazon AWS</p>
```

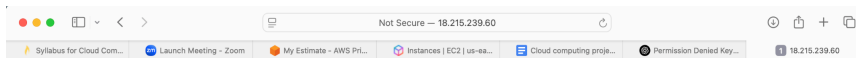
```
</body></html>
```

'Esc' then ':wq'

```
root@ip-172-31-40-251:/var/www/html# vi index.html
root@ip-172-31-40-251:/var/www/html# cat index.html
<html><body><h1>It works!</h1>
      <p>This website is hosted by Amazon AWS</p>
</body></html>

root@ip-172-31-40-251:/var/www/html#
```

Access the static web page by typing http://18.215.239.60/ into your browser:



It works!

This website is hosted by Amazon AWS

Run an Apache HTTP server benchmarking tool:

```
~$ sudo yum update
Last metadata expiration check: 1:26:52 ago on Fri 28 Feb 2025 02:16:35 PM UTC.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Upgrading:				
amazonlinux-rgo-cdn	noarch	2023.6.20250218-0.amzn2023	amazonlinux	16 k
crypto-policies	noarch	20240828-2.git626a059.amzn2023.0.1	amazonlinux	56 k
dnf	noarch	4.14.0-1.amzn2023.0.6	amazonlinux	459 k
dnf-data	noarch	4.14.0-1.amzn2023.0.6	amazonlinux	33 k
gnutils	x86_64	3.8.0-301.amzn2023.0.7	amazonlinux	1.1 M
openldap	x86_64	2.4.57-6.amzn2023.0.7	amazonlinux	256 k
openssl	x86_64	1:3.0.8-1.amzn2023.0.18	amazonlinux	1.2 M
openssl-libs	x86_64	1:3.0.8-1.amzn2023.0.18	amazonlinux	2.2 M
python3	x86_64	3.9.20-1.amzn2023.0.3	amazonlinux	28 k
python3-dnf	noarch	4.14.0-1.amzn2023.0.6	amazonlinux	409 k
python3-libs	x86_64	3.9.20-1.amzn2023.0.3	amazonlinux	7.5 M
rpm	x86_64	1:4.14.1.amzn2023.0.1	amazonlinux	3.4 M
system-release	noarch	2023.6.20250218-0.amzn2023	amazonlinux	28 k
yum	noarch	4.14.0-1.amzn2023.0.6	amazonlinux	30 k
Installing dependencies:				
dbus-libs	x86_64	1:1.12.28-1.amzn2023.0.1	amazonlinux	153 k
Installing weak dependencies:				
rpm-plugin-systemd-inhibit	x86_64	4.16.1.3-29.amzn2023.0.6	amazonlinux	18 k

```
Transaction Summary
Install 2 Packages
```

```

Total                                                                 1.8 MB/s | 325 kB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/4
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64 1/4
Installing : apr-util-1.6.3-1.amzn2023.0.1.x86_64 2/4
Installing : httpd-tools-2.4.62-1.amzn2023.x86_64 3/4
Running scriptlet: httpd-tools-2.4.62-1.amzn2023.x86_64 4/4
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/4
Verifying : apr-util-1.6.3-1.amzn2023.0.1.x86_64 2/4
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64 3/4
Verifying : httpd-tools-2.4.62-1.amzn2023.x86_64 4/4

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.1.x86_64      apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64      httpd-tools-2.4.62-1.amzn2023.x86_64

Complete!

```

```
~ $ ab -V
This is ApacheBench, Version 2.3 <$Revision: 1913912>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

The apache benchmark command `ab -c 50 -n 5000` sends -n=5000 total HTTP request to the specified URL `http://18.215.239.60/` -c=50 simultaneous requests at a time:

```
~ $ ab -c 50 -n 5000 http://18.215.239.60/
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking 18.215.239.60 (be patient)
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests
```

```
Server Software:      Apache/2.4.41
Server Hostname:      18.215.239.60
Server Port:          80
```

```
Document Path:        /
Document Length:      94 bytes
```

```
Concurrency Level:    50
Time taken for tests:  0.743 seconds
Complete requests:    5000
```

```
Concurrency Level:    50
Time taken for tests:  0.743 seconds
Complete requests:    5000
Failed requests:       0
Total transferred:    1815000 bytes
HTML transferred:     470000 bytes
Requests per second:  6727.08 [#/sec] (mean)
Time per request:     7.433 [ms] (mean)
Time per request:     0.149 [ms] (mean, across all concurrent requests)
Transfer rate:        2384.70 [Kbytes/sec] received
```

Connection Times (ms)

	min	mean[+/-sd]	median	max
Connect:	1	1 1.3	1	20
Processing:	1	6 3.3	5	29
Waiting:	1	6 3.0	5	29
Total:	2	7 3.8	6	33

Percentage of the requests served within a certain time (ms)

50%	6
66%	7
75%	8
80%	8
90%	12
95%	16
98%	20
99%	22
100%	33 (longest request)

```
~ $ █
```

-c=100 simultaneous requests at a time:

```
~ $ ab -c 100 -n 5000 http://18.215.239.60/
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

Benchmarking 18.215.239.60 (be patient)

```
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests
```

```
Server Software:      Apache/2.4.41
Server Hostname:      18.215.239.60
Server Port:          80
```

```
Document Path:        /
Document Length:      94 bytes
```

```
Concurrency Level:    100
Time taken for tests:  0.669 seconds
Complete requests:    5000
```

```
Concurrency Level:    100
Time taken for tests:  0.669 seconds
Complete requests:    5000
Failed requests:       0
Total transferred:    1815000 bytes
HTML transferred:     470000 bytes
Requests per second:  7469.31 [#/sec] (mean)
Time per request:     13.388 [ms] (mean)
Time per request:     0.134 [ms] (mean, across all concurrent requests)
Transfer rate:        2647.81 [Kbytes/sec] received
```

Connection Times (ms)

	min	max	mean[+/-sd]	median	max
Connect:	1	2	1.5	1	17
Processing:	2	11	4.0	11	32
Waiting:	1	11	3.7	10	30
Total:	7	13	3.7	12	34

Percentage of the requests served within a certain time (ms)

50%	12
66%	14
75%	15
80%	15
90%	18
95%	21
98%	25
99%	27
100%	34 (longest request)

```
~ $ █
```

-c=150 simultaneous requests at a time:

```
~ $ ab -c 150 -n 5000 http://18.215.239.60/
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

```
Benchmarking 18.215.239.60 (be patient)
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests
```

```
Server Software:      Apache/2.4.41
Server Hostname:      18.215.239.60
Server Port:          80
```

```
Document Path:        /
Document Length:       94 bytes
```

```
Concurrency Level:     150
Time taken for tests:   0.718 seconds
Complete requests:      5000
```

```
Concurrency Level:     150
Time taken for tests:   0.718 seconds
Complete requests:      5000
Failed requests:        0
Total transferred:      1815000 bytes
HTML transferred:       470000 bytes
Requests per second:    6963.70 [#/sec] (mean)
Time per request:       21.540 [ms] (mean)
Time per request:       0.144 [ms] (mean, across all concurrent requests)
Transfer rate:          2468.58 [Kbytes/sec] received
```

Connection Times (ms)

	min	mean[+/-sd]	median	max
Connect:	1	2 3.5	1	25
Processing:	2	19 6.6	17	49
Waiting:	2	18 6.0	16	45
Total:	8	21 6.8	18	49

Percentage of the requests served within a certain time (ms)

50%	18
66%	20
75%	23
80%	26
90%	32
95%	35
98%	41
99%	42
100%	49 (longest request)

```
~ $ █
```

-c=200 simultaneous requests at a time:


```
~ $ ab -c 200 -n 5000 http://18.215.239.60/
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

```
Benchmarking 18.215.239.60 (be patient)
```

```
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests
```

AWS CloudShell

```
Server Software:      Apache/2.4.41
Server Hostname:      18.215.239.60
Server Port:          80
```

```
Document Path:        /
Document Length:       94 bytes
```

```
Concurrency Level:     200
Time taken for tests:   0.656 seconds
Complete requests:     5000
```

```
Concurrency Level:     200
Time taken for tests:   0.656 seconds
Complete requests:     5000
Failed requests:        0
Total transferred:     1815000 bytes
HTML transferred:      470000 bytes
Requests per second:    7623.74 [#/sec] (mean)
Time per request:       26.234 [ms] (mean)
Time per request:       0.131 [ms] (mean, across all concurrent requests)
Transfer rate:          2702.56 [Kbytes/sec] received
```

AWS CloudShell

```
Connection Times (ms)
```

	min	mean[+/-sd]	median	max
Connect:	1	2 2.9	1	31
Processing:	3	23 9.1	21	51
Waiting:	1	22 8.0	21	50
Total:	10	26 8.3	23	52

```
Percentage of the requests served within a certain time (ms)
```

50%	23
66%	24
75%	27
80%	28
90%	42
95%	44
98%	50
99%	51
100%	52 (longest request)

```
~ $ █
```

-c=250 simultaneous requests at a time:

```
~ $ ab -c 250 -n 5000 http://18.215.239.60/
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

```
Benchmarking 18.215.239.60 (be patient)
```

```
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests
```

AWS CloudShell

```
Server Software:      Apache/2.4.41
Server Hostname:      18.215.239.60
Server Port:          80
```

```
Document Path:        /
Document Length:       94 bytes
```

```
Concurrency Level:     250
Time taken for tests:   0.715 seconds
Complete requests:     5000
```

```
Concurrency Level:     250
Time taken for tests:   0.715 seconds
Complete requests:     5000
Failed requests:        0
Total transferred:     1815000 bytes
HTML transferred:      470000 bytes
Requests per second:    6994.50 [#/sec] (mean)
Time per request:       35.742 [ms] (mean)
Time per request:       0.143 [ms] (mean, across all concurrent requests)
Transfer rate:          2479.50 [Kbytes/sec] received
```

AWS CloudShell

```
Connection Times (ms)
```

	min	max	mean[+/-sd]	median	max
Connect:	1	3	4.2	2	31
Processing:	4	31	10.0	30	63
Waiting:	1	29	9.7	28	63
Total:	20	34	9.6	32	66

```
Percentage of the requests served within a certain time (ms)
```

50%	32
66%	34
75%	37
80%	39
90%	48
95%	58
98%	62
99%	63
100%	66 (longest request)

```
~ $ █
```

Terminate Instance:

```

~ $ aws ec2 terminate-instances --instance-ids i-08c7ea551987fbc8f
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
:....skipping...
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
:....skipping...
{
  "TerminatingInstances": [

```

```

    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
:....skipping...
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
}

```

AWS CloudShell

```
~
(END)...skipping...
(END)...skipping...
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
~
(END)...skipping...
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
```

```
        "PreviousState": {
          "Code": 16,
          "Name": "running"
        }
      ]
    }
  ~
  ~
  ~
  (END)...skipping...
  {
    "TerminatingInstances": [
      {
        "InstanceId": "i-08c7ea551987fbc8f",
        "CurrentState": {
          "Code": 32,
          "Name": "shutting-down"
        },
        "PreviousState": {
          "Code": 16,
          "Name": "running"
        }
      }
    ]
  }
  ~
  ~
  ~
```

AWX CloudShell

(END)...skipping...

```
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
```

}

~

~

~

~

(END)...skipping...

```
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-08c7ea551987fbc8f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      }
    }
  ]
}
```

AWS CloudShe

```
        "PreviousState": {
          "Code": 16,
          "Name": "running"
        }
      ]
    }
  }
  ~
  ~
  ~
  ~
  ~
  ~
  ~
  (END)...skipping...
  {
    "TerminatingInstances": [
      {
        "InstanceId": "i-08c7ea551987fbc8f",
        "CurrentState": {
          "Code": 32,
          "Name": "shutting-down"
        },
        "PreviousState": {
          "Code": 16,
          "Name": "running"
        }
      }
    ]
  }
```

AWS CloudShell

S3

Create a bucket:

```
~ $ aws s3 mb s3://carrd12
make_bucket: carrd12
~ $
```

Verify bucket was created:

```
~ $ aws s3 ls
2025-02-24 13:27:24 carrd12
~ $
```

Use vi to create a simple html file called index.html. The content of the file is as follows:

Vi index.html to write the web file

600dd to delete the lines

Press 'i' for insert then enter the desired text

```
<html><body><h1>It works!</h1>
<p>This website is hosted by Amazon AWS</p>
</body></html>
```

'Esc' then ':wq'

```
~ $ vi index.html
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadGetObject",
            "Effect": "Allow",
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ]
        }
    ]
}

<html><body><h1>It works!</h1>
    <p>This website is hosted by Amazon AWS</p>
</body></html>
```

Upload the file into the bucket and make the file available to the public:

```
~ $ aws s3 cp index.html s3://carrd12
upload: ./index.html to s3://carrd12/index.html
~ $
```

Delete the public access block:

```
~ $ aws s3api delete-public-access-block --bucket carrd12
~ $
```

Set bucket policy to allow public read access. First, create a file called policy.json:

Vi policy.json to write the file

Press 'i' for insert then enter the desired text

```
{
```

```

"Version": "2012-10-17",
"Statement": [
  {
    "Sid": "PublicReadGetObject",
    "Effect": "Allow",
    "Principal": "*",
    "Action": [
      "s3:GetObject"
    ],
    "Resource": [
      "arn:aws:s3:::carrd12/*"
    ]
  }
]
}

```

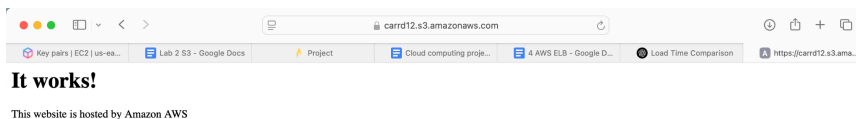
‘Esc’ then ‘:wq’

```

~ $ vi policy.json
~ $ cat policy.json
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::carrd12/*"
      ]
    }
  ]
}
~ $ 

```

Access the static web page by typing <https://carrd12.s3.amazonaws.com/index.html/> into your browser:



Run an Apache HTTP server benchmarking tool:

[CloudShell](#)
[Feedback](#)

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[Terms](#)
[Cookie preferences](#)

```

Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : apr-1.7.5-1.amzn2023.0.4.x86_64                  1/1
  Installing     : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64    1/4
  Installing     : apr-util-1.6.3-1.amzn2023.0.1.x86_64           2/4
  Installing     : httpd-tools-2.4.62-1.amzn2023.x86_64           3/4
  Running scriptlet: httpd-tools-2.4.62-1.amzn2023.x86_64           4/4
  Verifying      : apr-1.7.5-1.amzn2023.0.4.x86_64                  1/4
  Verifying      : apr-util-1.6.3-1.amzn2023.0.1.x86_64            2/4
  Verifying      : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64   3/4
  Verifying      : httpd-tools-2.4.62-1.amzn2023.x86_64           4/4

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64    apr-util-1.6.3-1.amzn2023.0.1.x86_64    apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64    httpd-tools-2.4.62-1.amzn2023.x86_64

Complete!
~ $

```

Verify ab -V tool is installed:

```

~ $ ab -V
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

~ $

```

The apache benchmark command `ab -c 50 -n 5000` sends `-n=5000` total HTTP request to the specified URL `http://carrd12.s3.amazonaws.com/` `-c=50` simultaneous requests at a time:

```

~ $ ab -c 50 -n 5000 https://carrd12.s3.amazonaws.com/index.html
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking carrd12.s3.amazonaws.com (be patient)
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests

Server Software:      AmazonS3
Server Hostname:      carrd12.s3.amazonaws.com
Server Port:          443
SSL/TLS Protocol:     TLSv1.3,TLS_AES_128_GCM_SHA256,2048,128
Server Temp Key:      X25519 253 bits
TLS Server Name:      carrd12.s3.amazonaws.com

Document Path:        /index.html
Document Length:      93 bytes

```

```

Concurrency Level:      50
Time taken for tests:    9.299 seconds
Complete requests:      5000
Failed requests:        0
Total transferred:      2625000 bytes
HTML transferred:       465000 bytes
Requests per second:    537.68 [#/sec] (mean)
Time per request:       92.992 [ms] (mean)
Time per request:       1.860 [ms] (mean, across all concurrent requests)
Transfer rate:          275.67 [Kbytes/sec] received

Connection Times (ms)
              min    mean[+/-sd] median    max
Connect:      7      40  19.1      34     199
Processing:   16      52  20.3      46     198
Waiting:      12      31  12.3      27     122
Total:        31      92  32.6      83     272

Percentage of the requests served within a certain time (ms)
 50%      83
 66%      93
 75%     105
 80%     111
 90%     137
 95%     164
 98%     190
 99%     209
100%     272 (longest request)

~ $

```

-c=100 simultaneous requests at a time:

```

~ $ ab -c 100 -n 5000 https://carrd12.s3.amazonaws.com/index.html
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking carrd12.s3.amazonaws.com (be patient)
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests

Server Software:        AmazonS3
Server Hostname:        carrd12.s3.amazonaws.com
Server Port:            443
SSL/TLS Protocol:       TLSv1.3,TLS_AES_128_GCM_SHA256,2048,128
Server Temp Key:         X25519 253 bits
TLS Server Name:         carrd12.s3.amazonaws.com

Document Path:          /index.html
Document Length:        93 bytes

```

```
Concurrency Level:      100
Time taken for tests:    9.132 seconds
Complete requests:      5000
Failed requests:         0
Total transferred:      2585000 bytes
HTML transferred:       465000 bytes
Requests per second:    547.55 [#/sec] (mean)
Time per request:       182.631 [ms] (mean)
Time per request:       1.826 [ms] (mean, across all concurrent requests)
Transfer rate:          276.45 [Kbytes/sec] received
```

Connection Times (ms)

	min	mean[+/-sd]	median	max
Connect:	9	85 27.4	78	175
Processing:	24	96 29.3	89	219
Waiting:	12	45 21.9	42	157
Total:	75	181 43.2	175	314

Percentage of the requests served within a certain time (ms)

50%	175
66%	196
75%	209
80%	219
90%	244
95%	259
98%	278
99%	290
100%	314 (longest request)

~ \$ █

-c=150 simultaneous requests at a time:

```
~ $ ab -c 150 -n 5000 https://carrd12.s3.amazonaws.com/index.html
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

Benchmarking carrd12.s3.amazonaws.com (be patient)

```
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests
```

```
Server Software:        AmazonS3
Server Hostname:        carrd12.s3.amazonaws.com
Server Port:            443
SSL/TLS Protocol:       TLSv1.3,TLS_AES_128_GCM_SHA256,2048,128
Server Temp Key:        X25519 253 bits
TLS Server Name:        carrd12.s3.amazonaws.com
```

```
Document Path:          /index.html
Document Length:        93 bytes
```

```

Concurrency Level:      150
Time taken for tests:   7.980 seconds
Complete requests:      5000
Failed requests:        0
Total transferred:      2525000 bytes
HTML transferred:       465000 bytes
Requests per second:    626.53 [#/sec] (mean)
Time per request:       239.415 [ms] (mean)
Time per request:       1.596 [ms] (mean, across all concurrent requests)
Transfer rate:          308.98 [Kbytes/sec] received

Connection Times (ms)
  min   mean[+/-sd] median   max
Connect:    5   116  32.9   113   225
Processing:  37  120  35.0   116   281
Waiting:    12   50  26.9    45   204
Total:      76  236  46.0   228  441

Percentage of the requests served within a certain time (ms)
 50%    228
 66%    250
 75%    260
 80%    271
 90%    295
 95%    326
 98%    355
 99%    390
100%   441 (longest request)
~ $ █

```

-c=200 simultaneous requests at a time:

```

~ $ ab -c 200 -n 5000 https://carrd12.s3.amazonaws.com/index.html
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking carrd12.s3.amazonaws.com (be patient)
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests

Server Software:      AmazonS3
Server Hostname:      carrd12.s3.amazonaws.com
Server Port:         443
SSL/TLS Protocol:     TLSv1.3,TLS_AES_128_GCM_SHA256,2048,128
Server Temp Key:      X25519 253 bits
TLS Server Name:      carrd12.s3.amazonaws.com

Document Path:        /index.html
Document Length:      93 bytes

```



```

Concurrency Level:      200
Time taken for tests:    8.300 seconds
Complete requests:      5000
Failed requests:         0
Total transferred:      2525000 bytes
HTML transferred:       465000 bytes
Requests per second:    602.40 [#/sec] (mean)
Time per request:       332.006 [ms] (mean)
Time per request:       1.660 [ms] (mean, across all concurrent requests)
Transfer rate:          297.08 [Kbytes/sec] received

Connection Times (ms)
      min    mean[+/-sd] median    max
Connect:    12   161   60.9    149    346
Processing:  53   163   61.3    153    451
Waiting:    13    80   49.4     68    269
Total:      136   324   81.1    307    747

Percentage of the requests served within a certain time (ms)
 50%    307
 66%    355
 75%    384
 80%    398
 90%    430
 95%    451
 98%    499
 99%    551
100%    747 (longest request)
~ $ █

```

-c=250 simultaneous requests at a time:

```

~ $ ab -c 250 -n 5000 https://carrd12.s3.amazonaws.com/index.html
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking carrd12.s3.amazonaws.com (be patient)
Completed 500 requests
Completed 1000 requests
Completed 1500 requests
Completed 2000 requests
Completed 2500 requests
Completed 3000 requests
Completed 3500 requests
Completed 4000 requests
Completed 4500 requests
Completed 5000 requests
Finished 5000 requests

Server Software:      AmazonS3
Server Hostname:      carrd12.s3.amazonaws.com
Server Port:          443
SSL/TLS Protocol:     TLSv1.3,TLS_AES_128_GCM_SHA256,2048,128
Server Temp Key:       X25519 253 bits
TLS Server Name:      carrd12.s3.amazonaws.com

Document Path:        /index.html
Document Length:      93 bytes

```

```

Concurrency Level:      250
Time taken for tests:   7.689 seconds
Complete requests:      5000
Failed requests:        0
Total transferred:      2525000 bytes
HTML transferred:       465000 bytes
Requests per second:    650.32 [#/sec] (mean)
Time per request:       384.429 [ms] (mean)
Time per request:       1.538 [ms] (mean, across all concurrent requests)
Transfer rate:          320.71 [Kbytes/sec] received

Connection Times (ms)
              min    mean[+/-sd] median   max
Connect:        3    186   64.3    181    343
Processing:     58    185   70.6    172    394
Waiting:        13     95   60.7     76    319
Total:         100    371   63.7    367    690

Percentage of the requests served within a certain time (ms)
 50%    367
 66%    391
 75%    398
 80%    414
 90%    436
 95%    485
 98%    551
 99%    625
100%    690 (longest request)
~ $ █

```

Remove an object, the index page:

```

~ $ aws s3 rm s3://carrd12/index.html
delete: s3://carrd12/index.html
~ $ █

```

Remove the bucket:

```

~ $ aws s3 rb s3://carrd12 --force
remove_bucket: carrd12
~ $ █

```

Data

EC2

URL	HTTP request	Time per request
http://18.215.239.60/	50	7.433 ms
http://18.215.239.60/	100	13.388 ms
http://18.215.239.60/	150	21.54 ms
http://18.215.239.60/	200	26.234 ms

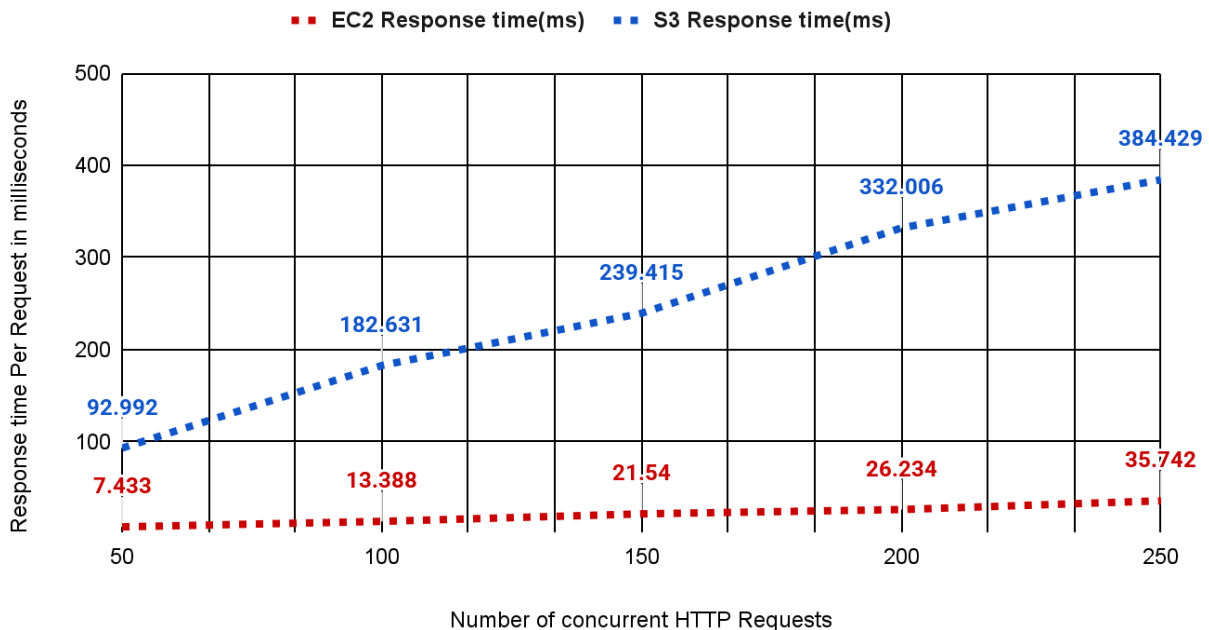
http://18.215.239.60/	250	35.742 ms
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S3

URL	HTTP request	Time per request
https://carrd12.s3.amazonaws.com/index.html/	50	92.992 ms
https://carrd12.s3.amazonaws.com/index.html/	100	182.631 ms
https://carrd12.s3.amazonaws.com/index.html/	150	239.415 ms
https://carrd12.s3.amazonaws.com/index.html/	200	332.006 ms
https://carrd12.s3.amazonaws.com/index.html/	250	384.429 ms

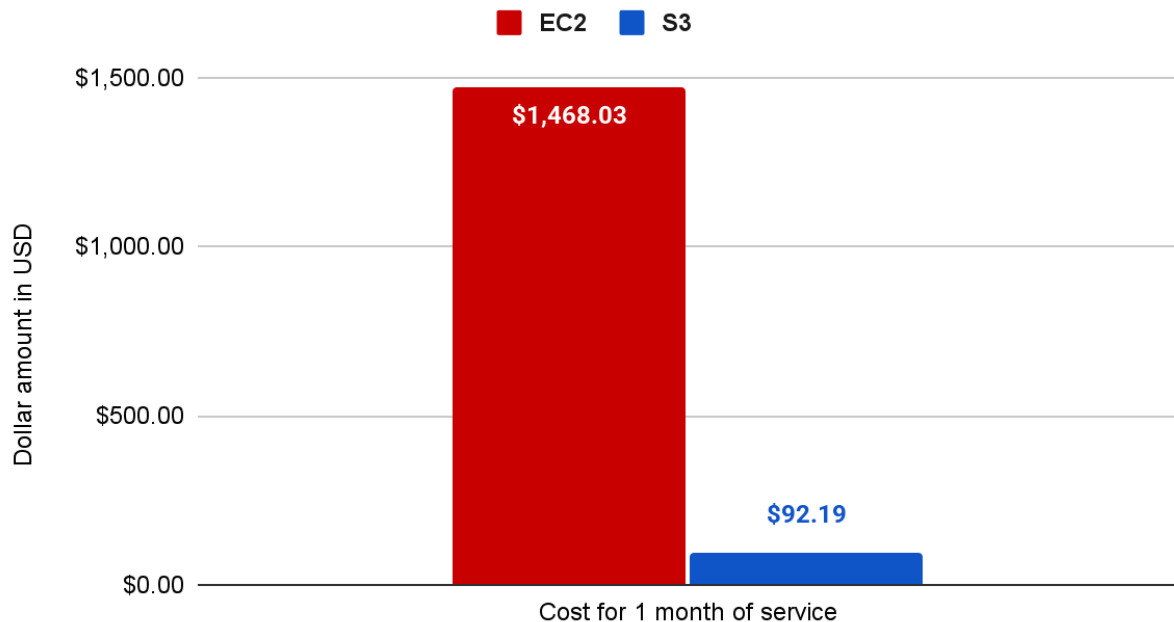
Analysis

Comparing the response time per request between EC2 and S3



Performance- When EC2 and S3 are compared in terms of performance, EC2 consistently performs better than S3, with noticeably faster response times as the number of concurrent requests rises. S3's response time increases exponentially, making it more than ten times slower at just 50 requests and reaching 384.429 ms with 250 requests, whereas EC2 can handle heavier loads with minimal latency escalation. This suggests that even while S3 is an affordable option for hosting static web pages, it has trouble with heavy traffic demands. On the other hand, EC2 is the recommended option for dynamic or heavily trafficked web applications due to its significantly greater performance and scalability.

Comparing the cost of service between EC2 and S3



Economic- Because Amazon S3 uses a pay-as-you-go model and only charges for storage and data transfer, it is a far more affordable alternative than EC2 for hosting a static website. On the other hand, EC2 charges more depending on the type of instance, uptime, and bandwidth used, particularly for instances that are operating continually. The trade-off for this cost benefit is that EC2 provides superior scalability and faster response times, while S3 is significantly slower under heavy traffic volumes. As a result, EC2 is more appropriate for high-performance applications, while S3 is best for static web pages with less traffic.

Conclusion

For this experiment, I contrasted the cost and performance of using EC2 with Amazon S3 to host a static website. According to my findings, S3 is more economical for sites with little traffic but suffers under high traffic volumes, with response times growing exponentially with increasing traffic. On the other hand, EC2 offers superior scalability and performance, managing more traffic with little increase in latency. Considering these results, I advise using S3 for static, low-traffic websites and EC2 for dynamic or high-traffic web applications.